

Research Master Thesis

Automatic Detection of Recurrent Gestures in Dutch

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Abstract

Recurrent gestures are conventionalized gestures with a stable relationship between their form and their meaning. One of these recurrent gestures is the palm up open hand gesture, which is typically made by extending the fingers to open the hand and rotating the lower arm so the palm of the hand is facing upwards. This gesture is used across many languages and cultures and is often made to present a point. However, there is still very little knowledge available on the different forms and functions of this gesture in Dutch. While automatic gesture recognition could decrease the time and manual labor needed for quantitative analyses of these gestures, which often rely on large annotated data sets, recognizing this gesture is difficult, partly due to the large amounts of variation present in gesture form. I compared the performance of two models: a static model that uses images as input and a dynamic model that uses video clips. I trained and tested both models with more and fewer form variants included in the data set. I found that the static model is less affected by the amount of variation in the form of the gesture and overall yields better results than the dynamic model. While further research is needed to develop models with better performances, this thesis illustrates some of the challenges that gesture recognition models face when recognizing palm up open hand gestures, and provides insights into the workings of these models that might improve their performance in future studies.

Declaration of Authorship

I, Fenna Berger, declare that this thesis, titled *Automatic Detection of Recurrent Gestures in Dutch* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Research Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 17/07/2026

Signed:



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Chapter 1

Introduction

Gestures are an important part of communication. They contain pragmatic and semantic information that adds an extra layer of meaning to spoken language. While gesturing is not mandatory for communication, it is found across cultures (Kendon 2004, Kita 2009), ages (Feyereisen and De Lannoy 1991) and communicative contexts (McNeill 1992, Chu et al. 2014). Gesturing is even found in children at the one word stage of language development (Iverson and Goldin-Meadow 2005) and in congenitally blind speakers talking to blind listeners (Iverson and Goldin-Meadow 1998, Chu et al. 2014).

There is a large variety of different types of gestures, all with their own form and function. Gestures can for example be *iconic*, which means the form of the gesture depicts something; *metaphoric*, representing an abstract concept or idea; *beats*, which are moving along with the rhythm of the speech; *cohesives*, connecting different parts of the discourse; or *deictic*, pointing to physical objects or abstract concepts previously mentioned in the conversation (McNeill 1992). Metaphoric gestures and iconic gestures can be collectively referred to using the term *representational gestures*. While McNeill's 1992 classification is one of the most widely used gesture classification systems, it does not include pragmatic gestures (for a system that does include this, see for example Kendon 2004).

One type of gestures, which has only more recently become a topic of interest in the field of gesture studies, is *recurrent gestures*. A gesture is a recurrent gesture if it "is used repeatedly in different contexts and its formational and semantic core remains stable across different contexts and speakers" (Müller 2010, p. 41). While there have been several qualitative studies on recurrent gestures (see for example Ladewig 2014 or Bressem and Müller 2014), quantitative studies are rare, due to the fact that these often need larger amounts of data that need to be annotated in some way to be able to draw statically significant conclusions. The process of annotating these data is difficult and costs a lot of time.

A possible solution for this problem would be to use automatic gesture recognition models to annotate the data. Gesture recognition models could identify potential interesting gestures that could then be labeled by expert human annotators. However, automatic gesture recognition models are difficult to train. This is partly due to the high amount of variation present within and across individuals making a certain gesture. Gesture recognition models need to be able to generalize across this variation to correctly classify gestures.

In this thesis, I investigated the broad research question *"How can gesture detection*

models recognize recurrent gestures in Dutch?". This question can be divided into two subquestions: (1) *"Which recurrent gestures exist in Dutch?"* and (2) *"How can gesture recognition models recognize these gestures?"*.

To answer this research question, I collected my own data in a small experiment. After a preliminary analysis of the data, I found that the palm up open hand gesture, a gesture where the speaker rotates their lower arm to present their palm to their interlocutor, often to present a point, was very common in the data, partly answering my first subquestion. This gesture has been discussed in detail in several qualitative analyses and has been shown to be used in both spoken and sign languages around the world (Müller 2004, Cooperrider et al. 2018). Finding this gesture in Dutch further contributes to the idea that palm up open hand gestures appear across languages and cultures. However, I was not able to find any existing research focusing on which types of palm up open hand gestures occur in Dutch. I made the decision to focus on this gesture specifically due to its large presence in the data set, the amount of literature that could be found on this topic in general, and the lack of literature on the forms this gesture takes in Dutch. This led me to the more specific research questions:

RQ1 (revised): *"What forms does the palm up open hand recurrent gesture take in Dutch?"*

RQ2 (revised): *"How can gesture recognition models recognize these gestures?"*
Subquestion: *"How do gesture recognition models deal with individual variation?"*
Subquestion: *"How do gesture recognition models deal with limited data?"*

The research in this thesis is very exploratory. Despite the fact that the palm up open hand gesture has been studied more than any other recurrent gesture, I was unable to find any studies that aim to automatically detect these gestures using machine learning. Because of this and because of the fact that I knew that I would be working with a small data set, my goal was not to develop the best performing model. Instead, I compared two different types of models with a focus on understanding how these models work, which factors can impact the performance of these models, and which challenges they face in recognizing palm up open hand gestures.

One of the factors I studied was the effect of variation in the form of the palm up open hand gesture, as is reflected in the first subquestion of RQ2. The different forms of the palm up open hand gesture are points on a continuum, ranging from a small finger lift to an extension of the entire arm (Cienki 2021). In order to recognize a palm up open hand gesture, a model needs to generalize across all these different gesture variants. By training and testing the models in conditions with varying amounts of variation in the data, I studied how the models deal with this variation in form.

Another factor is the limited availability of training data, which forms the second subquestion of RQ2. As described above, annotation involves a lot of manual labor, which costs time and money. This lack of training data often leads to class imbalance because there is more time the speaker is either not gesturing or making any other gesture than making a palm up open hand gesture. In these cases, a researcher needs to choose between having more training data, but a larger difference in number of samples per class, or using even less training data but having more balanced classes. This is made even more difficult when limiting the number of palm up open hand gestures based on the variation in form, as this further decreases the number of positive cases in the training data. I studied this trade-off between including more positive cases in the

training data and limiting the variation in the positive cases by selecting only specific palm up open hand variants.

In this thesis, I aimed to explore the challenges that arise when automatically recognizing palm up open hand gestures, as well as solutions for these issues which could be used in further gesture recognition studies. In addition to that, I created a small corpus of gestures in Dutch, annotated for different types of palm up open hand gestures. This corpus could be used for further research on palm up open hand gestures, or, if researchers are willing to add their own annotations, for other types of (gesture) research. The benefit of using this corpus as training data for the gesture recognition models is that the models are trained on naturalistic data. This is closer to the way language is used in everyday settings to communicate, and might include more variation than is the case for data that is created by gesturing in front of a camera in an artificial setting.

There are three main contributions of this thesis, I would like to highlight. Firstly, I created the annotated corpus I described above. Part of this corpus will be available to other researchers upon request. Which part this is, is based on and in line with the consent forms that were filled out by the participants. Secondly, in this corpus, I identified palm up open hand gestures in Dutch. I also differentiated between different gesture forms that were present, answering RQ1. Finally, I created two models for the automatic recognition of palm up open hand gestures and evaluated their performance across different conditions based on the amount of variation present in the data set as well as the size of the data set, answering RQ2 and making a methodological contribution to the field of gesture recognition.

This thesis is structured as follows: section 2 of this thesis discusses some existing literature on recurrent gestures, gesture recognition, and working with variation in gesture research. Section 3 focuses on the methodology used in data collection, annotation, model training, and evaluation. Section 4 consists of the results of all models. These results are discussed in section 5. Section 6 provides a summary and conclusion.

Chapter 2

Background

2.1 Recurrent Gestures

Recurrent gestures are conventionalized gestures with a stable relationship between form and meaning (Ladewig 2024). Although interest in recurrent gestures is a more recent development in the field of gesture studies and linguistics as a whole, recurrent gestures have long been studied in the fields of acting and rhetorics albeit under different names (Loeb 1912, Mosher 1916, Ott 1902). Ladewig hypothesized that this interest in recurrent gestures was due to their pragmatic function in natural conversation as well as their conventionalization which makes them easier to identify (Ladewig 2014).

When recurrent gestures were first discussed within linguistics, they were referred to by a large variety of names such as *pragmatic gestures* (Kendon 1995, Streeck 2005, Streeck 2009) or *interactive gesture*s (Bavelas et al. 1992, Bavelas et al. 1995). However, while these terms successfully capture the pragmatic function of these gestures, Ladewig (2014) stated that these terms do not capture the complete picture, as they fail to capture the language-like properties of these gestures. The term recurrent gesture was coined by Müller et al. during their project titled 'Towards a grammar of gesture: evolution, brain, and linguistic structures' which took place from 2006 to 2009. The handbook that was created based on this project was only published years later (Müller et al. 2013).

One of these properties of recurrent gestures is the fact that they are conventionalized and have a stable relationship between their form and meaning within a group of language users. Because of this, they show similarities to *emblems*, which also show a very stable relationship between form and meaning. Emblems serve the same function as verbal units and can therefore completely replace language. A waving gesture can, for example, serve as a way of greeting another person, and could therefore replace the phrase 'Hello'. However, this is not the case for recurrent gestures, which always act on or are part of a verbal utterance. An example of this would be a speaker making a pushing away gesture to express distance or disliking of the matter that is discussed. Here, the gesture is made in relation to what is being said. Because of this, recurrent gestures always contribute to a multimodal meaning of the utterance (Bressem and Müller 2014).

The function of a recurrent gesture influences the exact relationship between the gesture and the verbal utterance. Ladewig (2014) discusses two functions of recurrent gestures and their relation to speech. Firstly, she discusses recurrent gestures with a referential function. These are gestures that depict objects, actions, or events. In

this case, the gesture usually adds spatial information such as shape and or size of the depicted action, object, or event referred to in speech. If recurrent gestures have a referential function, they are co-expressed with a verbal unit. In her 2010 study on different types of recurrent gestures, Ladewig provides an example of this. In a German conversation about a movie, one of the speakers states that there were so many details that you can combine yourself. She makes a cycling gesture during the phrase that translates to *can combine yourself*. This is an example of a referential recurrent gesture, because it expresses that this combining is a mental process and therefore is used to depict a part of the action that is being described.

Secondly, recurrent gestures can fulfill a meta-communicative function and act as speech performatives (Teßendorf 2014). These gestures serve either a discursive function, acting upon the structure of the conversation and bringing attention to certain topics, or a modal function, expressing an attitude towards the matter that is being discussed. While co-expression with speech is still the most common, in this case recurrent gestures can occur in the pauses between different speech units. Especially recurrent gestures acting as performatives that aim to manage the behavior of other speech participants are quite often detached from speech.

Although the exact form of a recurrent gesture can vary depending on the context in which it is used, the main characteristics remain stable. This is called the *functional* or *kinesic core*. Recurrent gestures are often based on depictions of practical actions. This is called the *semiotic base*. While the form of the recurrent gestures might have started to differ from this original depiction during the conventionalization process, the semiotic base can still be derived from the functional core. Using this, the semantic core, which reflects the meaning of the gesture can be reconstructed. This is another aspect in which recurrent gestures differ from emblems, which are conventionalized to the extent that their from is no longer transparent.

2.1.1 Palm Up Open Hand Gestures

One particularly common type of recurrent gesture is the palm up open hand gesture. This gesture has two main characteristics: a rotation of the lower arm, causing the palm of the hand to face upward, and an extension of the fingers, causing the hand to be open. This gesture occurs frequently in both spoken and signed languages (for an overview, see Cooperrider et al. (2018)).

Although mentions of palm up open hand gestures go back as far as Quintilian (Loeb 1912), the first two authors discussing this from a linguistic point of view are Kendon (2004) and Müller (2004). Müller coined the name 'palm up open hand' in 2004. She described this gesture as metaphoric, stating that the form was rooted in the practical actions of giving and receiving. For both, she listed several senses the gesture could have. The giving-related senses of the palm up open hand were: (1) presenting an abstract object as visible or even obvious; (2) presenting an abstract object for joint inspection; and (3) proposing a shared perspective on an abstract object. The receiving-related senses all share the key idea that the palm is empty. These include: (1) to plead for an abstract object; (2) to request an abstract object; (3) to express openness to the reception of some abstract object; (4) to express the fact of not knowing (Müller 2004, as discussed in Cooperrider et al. 2018). These senses of palm up open hand gestures according to Müller as summarized in Table 2.1. The idea of the palm up open hand gesture being metaphoric was later followed by Streeck (2009). He introduced the concept of speech-handling. This describes how, what we now refer to

Type of gesture	Senses
Giving	Presenting an abstract object as visible or obvious Presenting an abstract object for joint inspection Proposing a shared perspective on an abstract object
Receiving	Pleading for an abstract object Requesting an abstract object Expressing openness to the reception of an abstract object Expressing the fact of not knowing

Table 2.1: The functions of giving- and receiving-related senses of palm up open hand gestures according to Müller.

as recurrent gestures, appear to act upon speech as if it were an object. One of the gestures he discussed in this chapter was the palm up open hand gesture.

Kendon (2004) referred to the palm up open hand gesture as the 'open hand supine'. He distinguished three types of open hand supine gestures: the *palm presentation gesture*, the *palm addressed gesture* (together these two were referred to as palm presenting gestures) and the *palm lateral gesture*. In a palm presentation gesture, the palm of the hand is displayed to the other interlocutor but not moved. Kendon states that this gesture can be used to (1) explain the meaning of a term or phrase; (2) expound the premises or conditions for understanding something; (3) give an account of circumstances that provide the setting of a narrative; or (4) give a conclusion or summary for something just said, or making a comment on it. The palm addressed gesture is similar to the palm presentation gesture, but here the hand is moved in the direction of another person or object. According to Kendon, this gesture can be used to (1) acknowledge something the other has said or to indicate it is correct or (2) to offer or receive something.

The palm lateral gesture is made by moving the open hand with the palm up laterally and often a bit backwards. While the two palm presenting gestures are used as an invitation or to convey offering or receiving, the palm lateral gesture indicates that the speaker is not going to take any action, due to an unwillingness or inability to do so. Kendon lists the following uses of palm lateral gestures: (1) unwillingness or inability to intervene in a situation; (2) something is declared obvious, nothing further can be added; (3) the speaker asks a rhetorical question; (4) the speaker responds to a proposal by neither accepting nor rejecting it; or (5) as an invitation. All uses of the palm up open hand gesture according to Kendon are summarized in Table 2.2

2.1.2 Quantitative Analyses of Palm Up Open Hand Gestures

Due to the high costs of manual annotation, quantitative analyses of palm up open hand gestures are very rare, though not non-existent. Here, I will briefly discuss two of these quantitative studies to give an idea of what has been done in this field.

The first study I will discuss is by Chu et al. (2014). This study focused on within group differences in gesturing by measuring correlations between a set of predictor variables and gesturing. Until then, correlational studies such as this one only considered *representational gestures*, which are gestures that use the hand shape and/or movement of the hand to depict an entity. One of the ways this study elaborates on the existing research is by adding *conduit gestures* (where the upwards facing palm of the hand is

Type of gesture	Senses
Palm Presentation	Explaining the meaning of a term or phrase Expounding the premises for understanding something Providing the setting for narrative Giving a conclusion or summary
Palm Addressed	Acknowledging something the other has said Offering to receive something
Palm Lateral	Unwillingness or inability to intervene Declaring something obvious Rhetorical question Neither rejecting nor accepting a proposal

Table 2.2: Types of palm up open hand gestures and their senses according to Kendon.

moved towards the listener, referred to as *conduit metaphor gesture* in McNeill (1992) or as a type of interactive gesture in Bavelas et al. (1992)) and *palm revealing gestures* (called palm lateral by Kendon) to their analysis. They state that different types of gestures serve different functions. Therefore, they could be related to a different set of cognitive or social skills. Chu et al. counted the number of gestures made while the participants performed two tasks: defining phrases and solving social dilemmas. The predictor variables were: verbal, visual, and spatial working memory capacity, empathy, spatial transformation and conceptualization abilities, and lexical retrieval ability. They found that of the total gestures made, 67% were representational gestures (with 6.05 gestures per 100 words), 15% were conduit gestures (with 1.47 gestures per 100 words), and 9% were palm revealing (with 0.80 gestures per 100 words), together making a total of 24% of palm up open hand gestures in their study. The number of palm up open hand gestures made by participants was particularly influenced by their empathy scores: participants with more empathy made more frequent and more salient conduit and palm revealing gestures. The authors state that this is likely due to the fact that palm up open hand gesture contributes to clear communication. Speakers with higher empathy might be more inclined to make sure the other is understanding what they are trying to convey and, therefore, might use palm up open hand gestures more frequently. They also found that conduit gestures and representational gestures are negatively correlated to visual working memory capacity, possibly due to the fact that these gestures might lighten some of the workload of the working memory by boosting the activation of the mental images. This study shows that there are several within group differences in gestures that are caused by differences in social and cognitive skills. Variation across, and even within individuals is, as will be discussed below, a major challenge in the field of gesture recognition.

The second quantitative study is Jehoul et al. (2017). This study focused on the palm up open hand gesture as part of a shrug. They describe the shrug as a compound enactment, existing of multiple gestures such as raising of the eyebrows or shoulders or tilting or shaking of the head. According to Jehoul et al, shrugs are most frequent in stance taking context and can, among other things, mean ignorance, indecisiveness or non-responsibility (Streeck 2009, Debras 2017). The data used in this study was the AliLeGra corpus (Feyaerts et al. 2012), which consists of roughly 8 hours of Flemish Dutch conversations between sets of two men who were under the impression that they were waiting for the actual experiment to begin. Out of the 1008 markers of

obviousness found by the researchers, the palm lateral gesture was the most common with 259 instances. These findings support Kendon in his statement that palm lateral gestures are used when the speaker claims or admits that something is obvious and nothing further can be done. The authors also connect their findings to Müller's ideas about inviting the interlocutor to share their opinion, claiming the speaker does not only want to mark the obviousness of the matter, but also wants to share this stance with their interlocutor.

2.1.3 Contribution to the Literature

In this thesis, I add to the existing literature by discussing recurrent gestures in Dutch. While these have been extensively discussed in German, I could not find any existing research discussing which recurrent gestures exist in Dutch. In this thesis, I focus on one type of recurrent gesture, the palm up open hand gesture. I discuss this gesture in more detail by showing what forms of this gesture occur in Dutch. However, the corpus I created could be used to study other recurrent gestures in Dutch in more detail. While not directly connecting palm up open hand gestures to predictors, the variation that is caused by these and other predictors is an essential part of this thesis, as will be discussed in Chapter 2.3.

2.2 Gesture Recognition and Classification

One of the reasons quantitative studies on gesture are so rare is the high cost of manual annotation. A possible solution for this is the use of automatic gesture recognition and/or classification models. In this section, I discuss some commonly used methods in this field as well as some common challenges in gesture recognition.

2.2.1 Common Techniques in Gesture Recognition

The field of gesture recognition focuses on automatically recognizing gestures, not only for linguistic research but also as a means of interaction between user and computational system (Xu 2017, Zou et al. 2018, Tateno et al. 2019, Ojeda-Castelo et al. 2022). Within this field a distinction between three types of gestures is made, depending on the part of the body these gestures are made with. Gestures can either be made with the face (Li and Zhang 2019), the body (Li et al. 2024), or the hands (Li et al. 2019). Because the focus of this thesis lies on hand gestures specifically, I will focus on these for this literature review. Another distinction that can be made is the distinction between vision-based models, which use video data or images as input, (Sinha et al. 2019) and sensor-based models, which use sensors, usually attached to gloves, which track the position of the hand and use these as input (Kim et al. 2019). Although sensor-based models have the advantage that they do not need to estimate the location of the different parts of the hand to track the hand position, but instead rely on more direct input features, wearing gloves with hand-tracking sensors is more invasive than using video data as input as the feeling of wearing gloves might alter the gesturing of the participant. Therefore, most of the existing research that is done on hand gesture recognition is done using video-based models, as is the case in this thesis (Ojeda-Castelo et al. 2022).

The final distinction that can be made in the field of gesture recognition is the distinction between static and dynamic gesture recognition. In static recognition, the

	Static models	Dynamic models
Input	Image or video	Video
Processes	Single frames	Windows with several frames
Recognizes	Poses	Movements
Common models	SVM, k-NN, K-means, mean shift clustering	DTW, HMM

Table 2.3: Comparison of characteristics of static and dynamic gesture recognition models.

models are trained to recognize poses while dynamic gesture recognition focuses on recognizing a movement (Al-Shamayleh et al. 2018). In this thesis, I compare a static gesture recognition model to a dynamic gesture recognition model. Therefore, I will now briefly discuss both types of models. For an overview of the characteristics of both types of models, see Table 2.3

Static gesture recognition is most commonly used in models that aim to recognize finger-spelled signs in sign languages. They are particularly suited for this due to the fact that most of the signs for letters are characterized by a specific hand shape, rather than a movement. Cheok et al. (2019) performed a systematic literature review of vision-based gesture recognition. They found that the most commonly used techniques for static gesture classification are support vector machine (SVM), k-NN, K-means and mean shift clustering, where SVM has been found to yield the best performance (see for example Tharwat et al. 2015 and Kurdyumov et al. 2011). Kurdyumov et al. (2011) were able to classify the finger-spelling signs in American Sign language with 92-93% accuracy using a Gaussian or linear kernel SVM.

Dynamic gesture recognition is more commonly used to determine if two different signs or gestures belong to the same class. Here, the most frequently used techniques are dynamic time warping (DTW) and Hidden Markov models (HMMs). In dynamic gesture recognition, two gestures cannot be compared frame by frame because the speed and length of the gestures might differ, even if the gestures belong to the same gesture type. DTW and HMM are both able to align the frames of the gesture that is made and then compute the likelihood of similarity. DTW compresses and expands the time of the gesture or sign to find the closest mapping between the two gestures. Darrell and Pentland (1993) used DTW to recognize the dynamic sign hello in a continuous phrase with four other signs. They achieved an average accuracy of 96%. First-order HMMs are based on two assumptions: (1) that the likelihood of the current state is only dependent on the previous state and (2) that the probability of the output that is produced by the current state is a direct result of this state. Chen et al. (2003) used HMMs to classify gestures into 20 different categories with 1200 train and test sequences. They achieved an accuracy of 98.5%.

2.2.2 Challenges in Gesture Recognition

Finally, I will discuss some common challenges in the field of gesture recognition.

Temporal Variance: One of these challenges, which was already briefly discussed above, is temporal variance. The same gesture can be made with large differences in the time used by the person making the gesture. This can be a result of differences in speeding up and slowing down the movement or overall speed. Gesture recognition

models need to be able to recognize that two gestures are similar, even if they look completely different when comparing them frame by frame. Possible solutions for this are the use of the techniques described in the previous section.

Spatial complexity: Another common challenge is the effect of the spatial complexity of gestures. The same gesture can be made bigger or smaller, in different places relative to the body of the speaker, and in some cases, a gesture can even be rotated. Gesture recognition models need to be able to generalize across all these different instances of the same gesture. This can for example be done by using features that are based on the shape of the hand that is making the gesture, which are more robust to differences in the location of the hand.

Camera angle: Another challenge that might be mediated by the use of features based on hand shape instead of hand position is the effects of differences in camera angle across recordings. This can have a major influence on how the gesture looks in the input data, so models need some way to estimate the 3D position of the hand. One solution is the use of multiple cameras to create a 3D image of the hand. However, this is more computationally expensive and requires more recording equipment and manual labor during data collection.

The use of multiple cameras can also decrease the chance that (part of) the hand is occluded in the recording. Within vision-based gesture detection models, it is hard to create a camera setup in which both hands are visible all the time. While a solution for this could be the use of gloves that can be tracked by sensors, this method of recording is more invasive than vision-based methods, as is described above.

Annotated data: Furthermore, while gesture recognition could potentially solve the problem of the amount of manual labor involved in annotation, the need for annotated data also forms a problem for developing the gesture recognition models. These models also require high quality annotated data for both training and testing. A potential solution would be to use machine learning techniques such as transfer learning, in which models are pre-trained on large datasets and then finetuned on a smaller dataset that is relevant for the task that needs to be performed, or unsupervised machine learning, in which no labels are provided but instead the data is grouped based on similarities.

Segmentation of the gestures: Finally, a challenge for gesture recognition is the segmentation of the gestures. In order to learn how to recognize and classify gestures, machine learning models need to be provided with positive and negative training instances. However, due to the fact that gestures often occur in sequences, can sometimes be cyclic or repeated, and not always have a clear end or beginning, annotating gestures is difficult to do in an objective manner. While a potential solution would be to produce gestures in isolation in an artificial setting, this would limit the model's ability to recognize gestures in natural data (Nam and Wohn 1996).

2.3 Variation in Gestures

2.3.1 Dealing with Variation in NLP Tasks

In both the fields of natural language processing (NLP) and theoretical linguistics, large amounts of variation in the data are often seen as noise. However, this variation is a core characteristic of language, as it is used by language users to express their identity by showing closeness to or distance from specific social groups (Curry et al. 2024). By

trying to eliminate this variation from our training data, we ignore an essential part of language that should be an integral part of any NLP research setup (Lutgen et al. 2026).

Schmidt (2010) uses the term *variation space* to refer to the space containing all different ways to produce language. He distinguishes three levels of synchronization that language users use to structure the variation space with. The smallest level is the micro-level: this level contains all individual interactions, such as a couple developing a private register to communicate in. The meso-level consists of larger groups of people such as social or regional groups who develop a shared dialect. The final level is the macro-level: this is used to describe standard languages. NLP research often only focuses on this macro-level and treats variation as disruptive (Plank 2016), ignoring all social meaning in variation.

Lutgen et al. (2026) describe how variation can cause challenges in different parts of language modeling and how these challenges can be tackled without discarding the variation in the data. I will highlight the two recommendations most relevant for this thesis here. Firstly, they emphasize the importance of knowing your data. Kreutzer et al. (2022) show that some of the most commonly used web-crawl corpora contain mistakes in the labeling of languages. This is even more often the case in low-resource languages, which are more frequently minority languages (Goot 2025). The impact of this is shown in Lau et al. (2025).

The second recommendation is to base your choices in data selection on which variety should be represented instead of by the amount of data available. They discuss Ramponi (2024) as an example, showing the shortcomings of NLP for non-standard varieties of Italian. Due to the focus on standard Italian, the cultural context of language is ignored. Ramponi argues for a speaker-centric approach that could preserve language varieties.

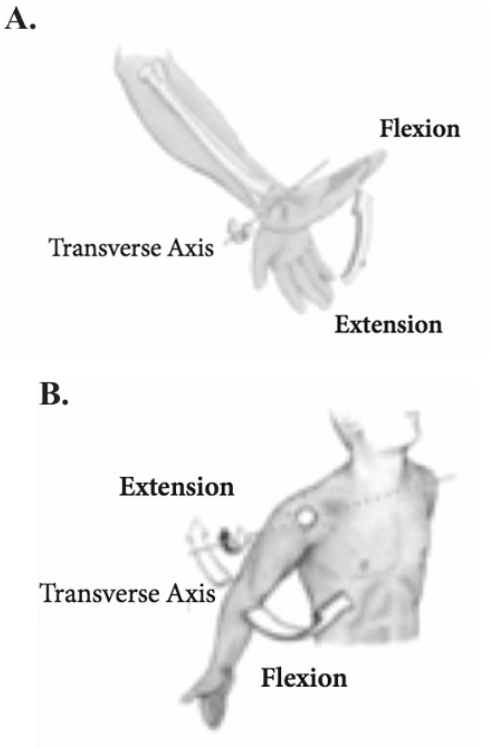
2.3.2 Variation in Form of the Palm Up Open Hand Gesture

In the case of the palm up open hand gesture, there is a lot of variation in form both across and within individuals. Cienki (2021) observes that there is not one single palm up open hand gesture but that there is a range of forms that can be described as a continuum. To illustrate this, he describes four points on this continuum, each representing a different form of the palm up open hand gesture.

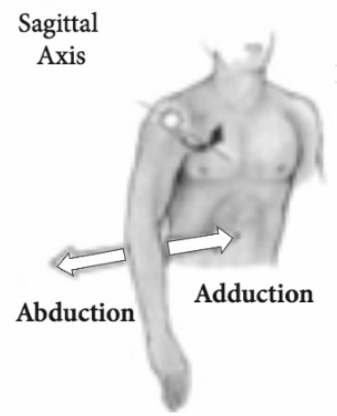
Cienki argues that the most commonly used system for describing gestures (developed by Stokoe Jr (1960/2005), widespread because of McNeill (1992), and further elaborated on by Bressem et al. (2013)) does not accurately capture the similarities between the different forms of the palm up open hand gesture. This is the case because this system, which focuses on hand shape, orientation of the palm of the hand, direction and manner of motion of the hands, and the location in which the gesture was produced in relation to the front of the speaker, uses external features to describe the gesture forms.

Instead, they propose that the kinesiological system developed by Boutet et al. (2018), which takes an internal perspective to form, would better highlight the connections between these different points on the continuum. This system looks at segments, which they define as "non-formable anatomical unit[s] situated between two articulations [joints]" (Boutet et al. (2018), p. 144) which can be moved in different manners. Each of those manners is called a degree of freedom. This system considers four types of movement, shown in Figure 2.1: (1) flexion, bending at the joint, versus extension,

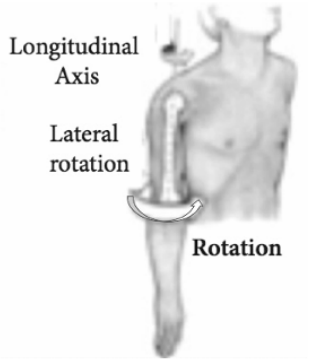
stretching at the joint; (2) abduction, movement away from the center of the body, versus adduction, movement towards the center of the body; (3) rotation; and (4) supination of the hand, where the palm is facing up, and pronation of the hand, where the palm is facing down.



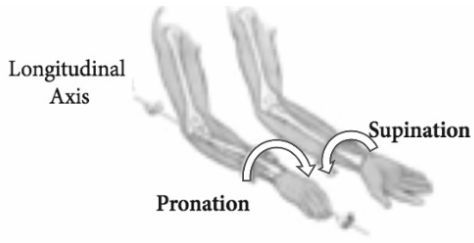
(a) Flexion and extension at the wrist (A) and elbow (B) (Boutet et al. 2018, p. 145).



(b) Abduction and Adduction from bending the elbow (Boutet et al. 2018, p. 145).



(c) Rotation of the upper arm (Boutet et al. 2018, p. 145).



(d) Pronation and supination of the hand (Boutet et al. 2018, p. 145).

Figure 2.1: Illustrations of the different manners in which segments could be moved in the kinesiological system (Boutet et al. 2016, taken from Boutet et al. 2018, p. 145).

To capture both the similarities, that show how these four gesture forms are part of the same gesture type, and the differences, that indicate why gesture recognition models might struggle classifying these forms as one, I will briefly describe the four gestures using both systems. I also provide an overview of these four gestures and their

descriptions in Table 2.4

The smallest version of the palm up open hand gesture is the finger lift. In the external system, this is described as a small lift of the thumb or fingers, most commonly made when the hands are placed on a table. It is worth noting that small movements of the fingers would normally be classified as a self-adapters, which are self-directed and often involve the speaker adjusting something on their person, for example their hair or clothing, and/or helping the speaker focus or calm themselves. However, the finger lift distinguishes itself from self-adapters because it is an outward movement instead of a movement that is self-oriented or self-touching. In the internal system, the finger lift is described as an extension of at least the joint that connects the finger to the hand, and possibly the other joints in the finger.

The next point on the continuum that was discussed is the hand turn-out, also referred to as the hand flip (Ferré 2011). In the external system, this is described as the hand turning out from the lower arm. However, it is not fully turned out and therefore does not reach a palm up position. In the internal system, the hand turn-out is described as a combination of the extension that was also present in the finger lift, and a partial rotation of the forearm from the elbow and wrist. This rotation is away from the body and can therefore be referred to as abduction.

The classic palm up open hand is described as an extension of the wrist, often combined with a slight lowering of the hand. The hand can move away from the body. This gesture can occur with either the palm facing up, or, more commonly, with the palm facing diagonally. In the internal system, this form is described as a larger extension of the fingers that leads to the opening of the hand. In addition, there is also a larger rotation of the lower arm, which leads to (partial) supination of the hand.

Finally, in the maximal version of the palm up open hand gesture, the hand is placed far from the body with a fully outstretched arm. The hand can either be located to the side or in front of the torso. In the internal system, this form is described as a combination of all of the movements described above and an extension at the elbow as well as an abduction of the shoulder.

Cienki states that using the internal perspective shows that the main difference between points on the continuum is which joints are involved in the movement and, as a result, which degrees of freedom are available.

2.3.3 Variation in This Thesis

In my thesis, I treated variation as an essential part of the data set I collected. Instead of decreasing the amount of variation that is present in the data by selecting only the instances that match the standard form, I annotated the gestures based on their form. Then, I evaluate how well the models I train handle variation in form between different instances of gestures of the same gesture type by comparing the performance of models trained on data with all variation present to models trained and tested on a more uniform data set.

In the next chapter I describe the process of data collection, annotation, as well as the models and conditions I used to test how gesture recognition models deal with limited data and variation in form of the palm up open hand gesture.

Gesture form	External description	Internal description
Finger lift	Small lift of thumb or fingers	Extension of joint that connects fingers to hand
Hand turn-out	Hand turning out partially from lower arm	Above + partial rotation of the forearm from the elbow/wrist
Classic palm up open hand	Extension of the wrist, combined with a slight lowering of the hand	Above + larger extension of the fingers, larger rotation of the arm, partial supination of the hand
Maximal palm up open hand	Hand is far from the body with a fully outstretched arm	Above + extension at the elbow, abduction of the shoulder

Table 2.4: Description of four forms of the palm up open hand gesture in the external and internal systems, adapted from Cienki (2021).

Chapter 3

Methodology

3.1 Data Collection

The data collection process follows the methodology used in Cienki et al. (Forthcoming) (briefly described in Cienki et al. (2025)and Prokofyeva et al. (2025)). In order to ensure that this project complies with the Ethical Guidelines of the Vrije Universiteit, this project was submitted for review by the Research Ethics Review Committee and Research Integrity [1]. The project was approved on 23/04/2026.

3.1.1 Recording setup

Data collection took place over three days in May 2026. I have a included a schematic representation of the setup in Figure 3.1.I made all recordings in the classrooms in the VU Main Building. In these rooms, I set up two chairs in front of a large wall with a neutral color in such a way that the chairs were facing each other at a 90 degree angle. In front of the chairs was a table with a laptop on top. I placed the table in such a way that participants could just reach the laptop without standing up from the chairs. To make the video recordings, I used three cameras: one to film both participants in the same frame and one individual camera for each of the participants. All cameras I used were GoPros, two of them were the HERO12 Black model, and one was the HERO9 Black. I placed the overview camera on the table behind the laptop. The two individual cameras were placed in such a way that they were facing the participants. To record the audio, I gave each of the participants a lapel microphone. The microphones I used were the Boya MIC2-02.

3.1.2 Participants

I recorded conversations between a total of ten pairs of two participants (N=20, of which 13 were female). All participants were native speakers of Dutch and each pair of participants consisted of two students who already knew each other prior to the data collection session. This ensured that the conversation would not be awkward or difficult because the participants felt uncomfortable with each other. The participants were compensated for their time with a 15 euro VVV gift card. I recruited the participants by asking students who happened to be on the VU campus on the days the data

[1] https://vu.nl/en/about-vu/faculties/school-of-humanities/more-about/research-ethics-review-committee-and-research-integrity, accessed on 09/07/2026

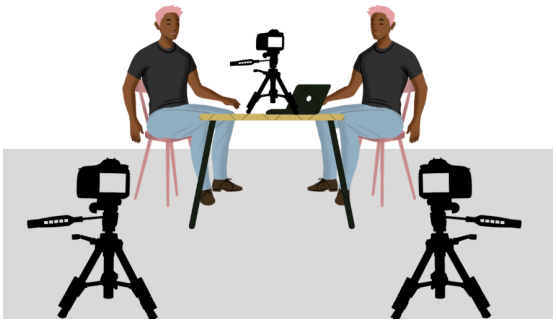


Figure 3.1: Schematic representation of the setup that I used for data collection.

collection took place if they had time to participate in my data collection. Because of this, the participants came from a wide range of different study programs. However, the demographic of this group is still limited in terms of level of education and age range.

3.1.3 Method

During the data collection, I first asked the participants to sit at a table where I had previously laid out the participant information document containing information about the data collection and the consent form. In the participant information document, I explained that my data collection focuses on communication between two participants. In order to prevent them from focusing on their gestures during the conversation, I did not mention anything about gestures during the data collection. These forms are included in Appendix A.

After they had signed the consent forms, I asked the participants to move to the chairs where the cameras were set up and gave each of them a microphone to clip to their clothing. In order to ensure that all participants had something to discuss, I first showed them a video about artificial intelligence on the laptop that was placed on the table. Because this video was originally in English, I recorded a Dutch voiceover [2]. I explained the goal of showing them the video and emphasized that they would not be tested on the content of the video.

After the video had ended, I explained to the participants that the rest of the slides contained questions that they could discuss. I asked them to read the question, click to the next slide, which contained a black screen, and then discuss the question for two or three minutes. I told them not to worry about time too much and to focus on just having a conversation at their own pace.

After 20 minutes had passed, or after the participants had discussed all questions I told them the data collection was finished. I gave them a final consent form where they could indicate if they allowed using (anonymized) parts of the collected data in talks, publications, and/or on websites. This form is also included in Appendix A. I did not debrief participants about the goal of the data collection because I was still looking for additional participants during the first days of data collection and I hoped some of the participants would share this with their friends. While no participants were recruited

[2] `https://youtu.be/RzkD_rTEBYs?si=CJDYGgrmMN8V9O6s`. The version of the video with the Dutch voiceover, as well as the other materials used during data collection, can be found in Yoda.

this way, it could have been the case that they accidentally gave away the purpose of the data collection.

3.1.4 Initial Observations from Data Collection

Two of the questions that participants were asked to discuss contained the term *neuraal netwerk* (neural network). Several of the participants did not know what this term meant. Although all participants found a way to keep talking (for example, by making up a definition for the term or by skipping the question), it did cause confusion and brought the attention of the participants back to the fact that they were participating in a data collection session and not just having a natural conversation. To combat this, I told the last two duos to skip these questions if they were unsure what a neural network was.

There were two duos of participants whose conversation I had to interrupt due to something that happened between the participants. In one conversation, one of the participants took out their phone to fact check something that caused disagreement during the conversation. Because holding a phone would limit their ability to produce natural gestures, I interrupted and asked them to put their phone away until after the data collection had ended. In the second duo, one of the participants was convinced there was a secret hidden purpose to the data collection. Despite having explained multiple times that the focus of the data collection was really the conversation, and that it would be wrong for me to lie to them, they still got distracted by this to the point where it influenced the naturalness of the conversation. In order to point their attention back to the conversations, I assured them once more that the focus of the data collection was their conversation.

There were some technical malfunctions that affected the data collection. Firstly, one of the two individual cameras stopped recording in the middle of a recording session. The camera message showed that this was due to the SD card being full, but when checking the card later, there was still enough storage space left. I unfortunately did not notice that the camera stopped recording until after the conversation ended. Because of this, roughly half of that conversation is missing. Secondly, one of the screws on the stand of one of the individual cameras came loose, causing the camera to tilt down during the conversation that were recorded on the first day. Finally, I had to briefly interrupt one conversation to plug in the charger of the laptop that showed the slides.

3.2 Data

3.2.1 Preprocessing

I used Adobe Premiere Pro to synchronize the videos made by the three cameras and the two audio recordings from the lapel microphones. Then, I created one composite video for each data collection session so I could see all videos at the same time. I annotated the gestures in ELAN (Max Planck Institute for Psycholinguistics, The Language Archive 2026). For a detailed overview of how I made the annotations, see Appendix B. If possible, I always annotated the complete gesture, including stroke, hold, and retraction (Kendon 1980, McNeill 1992). In cases where this was not possible, the annotation only included the stroke and (part of) the hold.

I categorized the annotated data into two classes with four subclasses each. The first class, the positive class, contained all gestures that were some form of a PUOH gesture and the second class, the negative class, contained all (non-)gestures that were not. Within the positive class, I created four subclasses based on the gesture variants discussed in Cienki et al. (2025): thumb lift, hand turnout, open hand diagonal and palm up open hand. However, as these four gesture variants represent a continuum of palm up open hand gesture variants instead of discrete categories, and which label best suits a specific gesture depends on multiple variables (e.g., wrist rotation or extension of the fingers), dividing all gestures into one of these four categories can be challenging. In order to create simpler and more objective annotation guidelines, I only considered which parts of the arm and hand were involved in making the gesture when assigning the subclass label. I found that this was easy to see in most of the videos and often correlated with the other characteristics described in Cienki et al. (2025). This resulted in the following four subclass labels, images of which are shown in Figure 3.2:

- Thumb lift (TL): the gesture includes only movement of the thumb(s);
- Hand turnout (HT): the gesture includes a rotation of the wrist;
- Open hand diagonal (OHD): the gesture includes a rotation and/or extension at the elbow;
- Palm up open hand (PUOH): the gesture includes a rotation and/or extension at the shoulder.

The negative class, contained four subclasses, examples of which are shown in Figure 3.3:

- Other gesture (OG): any gesture that is not a form of the PUOH gesture;
- Self-adapter (SA): gestures such as fidgeting, scratching a body part or adjusting one's hair;
- No gesture (NG): parts of the data collection where the participant is not gesturing;
- Unsure (US): this class contains all gestures that look like a PUOH gesture in form, but, according to my intuitions as a native speaker of Dutch, were used in a different manner than is usually the case for a PUOH gesture (such as iconic).

Some participants were fidgeting throughout the whole conversation. In these cases, I annotated all of the larger *self-adapter* gestures and only a selection of the small fidgeting gestures. I made the *no gesture* annotations by randomly selecting parts of the video that were not already assigned any other annotation label and assigning a *no gesture* label. For each participant, I created around 15 *no gesture* annotations of around 1-2 seconds. All gestures where the hand making the gesture was not entirely visible throughout the duration of the whole gesture were excluded.

For the dynamic model, I used the annotations to extract video clips from the video data I had previously collected. The overview camera that filmed both participants was able to capture some gestures that the individual cameras did not capture because participants often had their hands in their lap, and the overview camera was placed higher and thus had a better angle. I created an individual video for each participant



(a) Example of a participant making a thumb lift (TL) gesture.



(b) Example of a participant making a hand turn-out (HT) gesture.

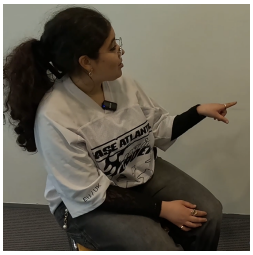


(c) Example of a participant making an open hand diagonal (OHD) gesture.



(d) Example of a participant making a palm up open hand (PUOH) gesture.

Figure 3.2: Examples of the four subclasses of the positive class in my data set.



(a) Example of a participant making an other gesture (OG).



(b) Example of a participant making a self-adapter (SA) gesture.



(c) Example of a participant making no gesture (NG).



(d) Example of a participant making an unsure (US) gesture.

Figure 3.3: Examples of the four subclasses of the negative class in my data set.

	Min	Max	Average	Total
Recording length	10:20	20:19	18:28	03:04:40
Number of gestures	45	288	181	1812
Number of positive gestures	4	135	57	567
Number of negative gestures	41	190	125	1245
Length of positive clips in minutes	00:00.30	00:09.11	0:01.58	14:56.53
Length of negative clips in minutes	00:00.30	00:09.14	0:01.90	39:28.63

Table 3.1: Minimum, maximum, average, and total recording length, number of total, positive, and negative gestures, and length of positive and negative clips.

by trimming 45-48% of the video side where the other participant was sitting. In some cases, it was impossible to do this in such a way that the gestures of the other participant were never visible without trimming off some of the gestures of the main participant in that video. In these cases, I attempted to find a middle ground between these two requirements.

The dynamic model uses separate features for the left and right hand. In order to prevent the data from being skewed by participants gesturing more with the hand that is closer to the other participant, which might be the case because palm presenting gestures are often made towards the interlocutor, I mirrored all videos of the right participants.

To create the images that were used as input for the static model, I manually took screenshots of the video clips. When possible, I took this screenshot of the hold of the gesture. However, in some cases, especially in the negative class, there was no real hold. In these cases, I selected a random part of the gesture in which the hands were as much in focus as possible.

All annotations, composite videos, individual participant videos, as well as the extracted images and video clips per gesture are stored in the project drive in Yoda, the cloud storage system recommend by the ethics committee.

3.2.2 Data exploration

Some statistics about the corpus are shown in Table 3.1. In total, the full videos had a duration of 3 hours, 4 minutes, and 40 seconds. The shortest video was that of session 2, which was only 10 minutes and 20 seconds long, due to the technical malfunction described above. Other than that, all videos fell within a range of 17 minutes and 50 seconds and 20 minutes and 19 seconds. The positive class video clips have a total duration of 14 minutes and 57 seconds and an average duration of 1.58 seconds. The video clips in the negative class had a total duration of 39 minutes and 29 seconds. Their average duration was 1.90 seconds, which is longer than that of the positive gestures.

Table 3.2 shows the total number of occurrences for each of the classes and subclasses. This table shows that the negative class, which has a total of 1245 instances, is more common than the positive class, which has 567 instances across the data set. Within the positive class, the most common gesture variant was the open hand diagonal (298 instances), and the least common was the thumb lift (12 instances). In the negative class, the most common subclass was the *Other gesture* subclass, which had a total of 580 instances across all videos. The least common was the *Unsure* category,

Class	Subclass	Total
Positive	Thumb lift	12
	Hand turnout	114
	Open hand diagonal	298
	Palm up open hand	143
	Total	567
Negative	Other gesture	580
	Self-adapter	247
	No gesture	279
	Unsure	139
	Total	1245
Overall	Total	1812

Table 3.2: Total number of occurrences of classes and subclasses of palm up open hand (PUOH) gestures. Positive indicates affirmative PUOH gesture; negative indicates another gesture type.

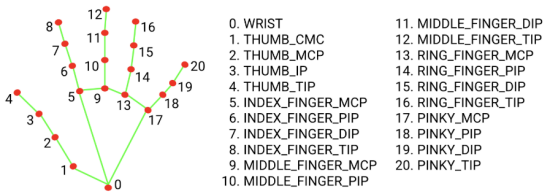


Figure 3.4: Overview of the 21 landmarks that are tracked by MediaPipe handtracking, taken from `https://developers.google.com/edge/mediapipe/solutions/vision/hand_landmarker`

which had a total of 139 instances.

Table 3.3 shows the variation in the number of gestures per participant. Excluding session 2, the total number of gestures made by a participant ranges between 22 and 197, with an average of 98.17. The average number of gestures in the positive class per person (excluding session 2) was 31.28 and the average number of negative class gestures per person was 66.89. The ratio of positive class gestures compared to total gestures per person ranges between 0.05 and 0.50 with an average of 0.29.

There is a lot of variation across individuals, both in the total number of gestures performed and in the ratio of positive gestures. On top of that, there is also a lot of variation across the classes and subclasses, with some being far more common than others. While this could be considered noise, this is a reflection of real life: language users each have their own way of gesturing and some gestures are more common than others. Therefore, I decided not to exclude anything from my data set.

3.3 Models

All code used in this thesis can be found on https://github.com/fennaberger/recurrent-gestures.

Participant	Positive	Negative	Total	Ratio Positive	Total gestures per minute
1L	3	22	25	0.12	1.24
1R	76	61	137	0.55	6.80
2L	**2**	**20**	**22**	**0.09**	2.06
2R	**2**	21	23	**0.09**	2.16
3L	24	**127**	151	0.16	7.71
3R	60	43	103	**0.58**	5.26
4L	**2**	37	39	0.05	2.19
4R	10	38	48	0.21	2.69
5L	22	91	113	0.19	5.66
5R	27	75	102	0.26	5.11
6L	18	59	77	0.23	3.95
6R	49	98	147	0.33	7.54
7L	**98**	95	**197**	0.50	9.70
7R	37	54	91	0.41	4.48
8L	16	30	46	0.35	2.55
8R	46	87	133	0.35	7.37
9L	16	83	99	0.16	5.02
9R	16	107	123	0.13	6.23
10L	24	55	79	0.30	4.11
10R	15	46	61	0.25	3.17
Average (excl 2)	31.28	66.89	98.17	0.29	5.04

Table 3.3: Total number of positive (PUOH) and negative (other) gestures per participant as well as the ratio of positive gestures made by each participant.

3.3.1 Static Model

The static model I adapted for this thesis is based on Tokhirov (2023). Tokhirov used his own webcam to create training data by holding his hand in different positions and pressing keys corresponding to the four categories he created to take pictures of this. Figure 3.5 shows each of the four categories with an example image. One of the categories, the *Hand Open* category, has the same hand shape as a full palm up open hand gesture. However, the palm of the hand is facing the camera instead of facing up. Nevertheless, the fact that the model was able to recognize this hand shape suggests it might also perform well when recognizing palm up open hand gestures. This resulted in very clear, high quality training data where the background was the same in all images and the location of the hand was always the same relative to the camera. Tokhirov used Mediapipe to determine the X and Y coordinates of 21 joints of the hand and wrist, as is shown in Figure 3.4. Using arrays of these landmarks as input he trained and tested a neural network, an XGBoost, and a random forest model to see which performed best.

The neural network that Tokhirov tested was a dense neural network. This means that each neurons in a layer, are connected to all neurons in the previous layer. During training, the connections between the neurons are adjusted to recognize patterns in the landmark data and to ultimately predict the correct gesture type.

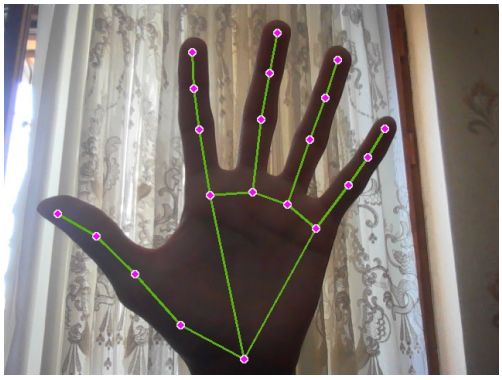
XGBoost (eXtensive Gradient Boosting) is an algorithm where multiple decision trees are sequentially trained to collectively form a stronger model. In this case, the first decision tree is trained to predict the gesture type based on the landmarks. After that, each decision tree is trained to predict and correct the errors of the previous trees. A learning rate is used to decrease the effect each new model has on the final prediction, leading to a more gradual learning process and a smaller risk of overfitting.

A random forest algorithm also uses multiple decision trees. However, here, the trees are trained in parallel, each one of them taking the landmarks as input and predicting the gesture type. Then, all predictions from the separate are combined, usually through majority voting, to end up with one prediction for each gesture in the data.

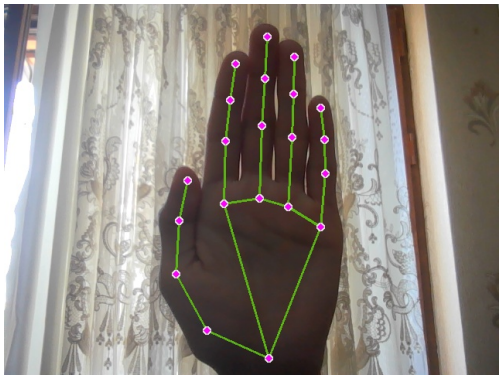
While Tokhirov did not report any quantitative descriptions of his results, he stated that the XGBoost model yielded poor results, even after retaking the pictures with improved lighting. The neural network and the random forest model performed better, predicting the correct hand position most of the time. There was no large difference in performance between these two models, and because the random forest model was faster, this was the model he implemented in his final hand recognition program.

For this thesis, I did not make many changes to Tokhirov's design as the objective is very similar. However, I made some small changes to be able to use existing images as input, instead of the webcam images Tokhirov used, and only used the random forest model to perform a binary classification.

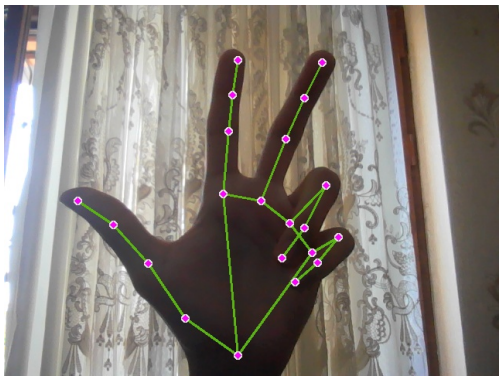
The biggest difference between this experiment and the original experiment performed by Tokhirov will be the image quality. Tokhirov was able to produce high quality images of hand positions because these were made in an artificial setting. This experiment uses more natural data, which means that there is variation in the form of the gesture even within the four subclasses of positive palm up open hand gestures. Furthermore, the gestures were made by 20 different speakers leading to even more variation. Finally, there is variation in the position of the hand relative to the camera and lighting of the room. Although it is difficult to compare the performance of the model without quantitative metrics describing the results, investigating how the model



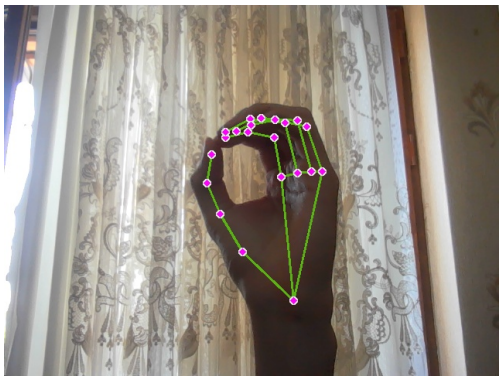
(a) Hand Open



(b) Hand Closed



(c) Hand Three



(d) Hand Zero

Figure 3.5: Images of the four categories used by Tokhirov including the 21 joints in the hand and wrist that were used for tracking.

performs on more natural data could still yield useful insights in gesture detection with limited data.

3.3.2 Dynamic Model

The design of the dynamic model I used is based on Pouw (2024) which was in turn based on Yung (2022). This is a convolutional neural network (CNN) that uses MediaPipe Holistic to extract landmarks from hand, body and face. Like the static model, this pipeline is designed to classify different types of gestures. In the case of the example data provided with the model, these were nodding, head turning and clapping. The model first predicts that probability that there is movement in each 25-frame window in the video clip. If this is the case, the model tries to predict the probabilities of each of these gesture types for that same window. To do so, MediaPipe extracts 29 features, 16 of which are related to hand position, which are used as input for the model:

- rotation_x (pitch) (in degrees)
- rotation_y (yaw) (in degrees)
- rotation_z (roll) (in degrees)
- nose_x (position in pixel)

- nose y (position in pixel)
- nose z (position in pixel)
- distance nose to chin (normalizer) (in normalized units)
- distance left eye brow to left eye center (in normalized units)
- distance right eye brow to right eye center (in normalized units)
- distance mouth corners (in normalized units)
- distance mouth apperture (in normalized units)
- distance left wrist to right wrist (in normalized units)
- distance left elbow to right elbow (in normalized units)
- distance left elbow to shoulder center (in normalized units)
- distance right elbow to shoulder center (in normalized units)
- distance left wrist to shoulder center (in normalized units)
- distance right wrist to shoulder center (in normalized units)
- distance left shoulder to left ear (in normalized units)
- distance right shoulder to right ear (in normalized units)
- distance left thumb to left index (in normalized units)
- distance right thumb to right index (in normalized units)
- distance left thumb to left pinky (in normalized units)
- distance right thumb to right pinky (in normalized units)
- distance x left wrist to x left elbow (in normalized units)
- distance x right wrist to x right elbow (in normalized units)
- distance y left wrist to y left elbow (in normalized units)
- distance y right wrist to y right elbow (in normalized units)
- distance left index finger to nose (in normalized units)
- distance right index finger to nose (in normalized units)

While some of these features are related to rotation, the hand position features are all based on the distance between two body parts. These distances are standardized against constant distances such as those from the chin to the top of the head or from left eye to right eye. In addition to these features, their time derivatives and standard deviations are used to capture the changes in these features over time.

A CNN consists of several connected convolutional layers, each with a set of filters. The model that I used has three layers, the first with 32 filters, the second with 64

filters, and the third with 128 filters. Each filter is applied to a small window that slides over the full 25-frame window that is being processed to recognize patterns. The output of a filter is called a feature map. After each convolutional layer, pooling is used to reduce the dimensionality of the feature maps before they are passed to the next layer. Finally, the output layer of the model predicts the probability that there is motion in the window, and the probabilities of each of the gesture types. The model was trained for 500 epochs with a patience of 10, meaning that the model was trained and evaluated on part of the training data for a maximum of 500 times. After each epoch, the training parameters were adjusted to further improve the model. However, if the model did not improve after 10 consecutive epochs, the training process was stopped and the current best model was used as the final model.

In order to convert the probabilities into labels, Pouw used thresholds. The thresholds for containing motion and containing the gesture that the model was meant to detect were set at probabilities of 0.5 and 0.9 respectively. If the threshold for motion was met, and one or more of the thresholds for the gesture types were met, the label of the gesture type with the highest probability was assigned. If the threshold for motion was not met, or the threshold for motion was met but none of the gesture type thresholds were met, the no motion label was assigned.

I made two changes to the original design of the CNN model. The effects of these changes is discussed in Chapter 5. The first is that I set the number of epochs to 5. Due to limited time and computational power, I was unable to train the model for the full number of epochs. The second change I made was necessary because I needed one label per video, instead of one per window. Because only one gesture needs to be made for the entire video to be classified as that gesture, I selected the highest probabilities for both motion and palm up open hand gestures and used those to determine the label for the full video clip.

3.4 Experiment

I trained both the static and the dynamic model with a 70/30 train/test split that was balanced across all eight subclasses. What differs between conditions are the subclasses included in the data that are used to train and test the models. Because there is such a large variation in form across different instances of the same gesture, I used different combinations of subclasses to test how increasing the variation within the positive class affect the models' ability to distinguish positive class gestures from negative class gestures.

In all conditions, the models were evaluated using precision, recall, F1 score, and accuracy. Accuracy expresses how many percent of the instances in a test set are labeled correctly by a model. However, in cases of class imbalance, accuracy scores can easily get very high by simply predicting the majority class for each instance, it is common to also look at the precision, recall and F1 score of a model's output. To understand these terms it is helpful to understand the terms shown in Figure 3.6. This type of image is called a confusion matrix, and it is useful to understand which of the classes are easy to predict and which are easily confused with others. The rows show the actual correct labels, in this example divided into two classes, positive and negative. The columns show the predictions made by the model. This results in four different categories. The first is the category of *true positives*, these are actually part of the positive class and correctly labeled by the model. The second category is the *false positives* category.

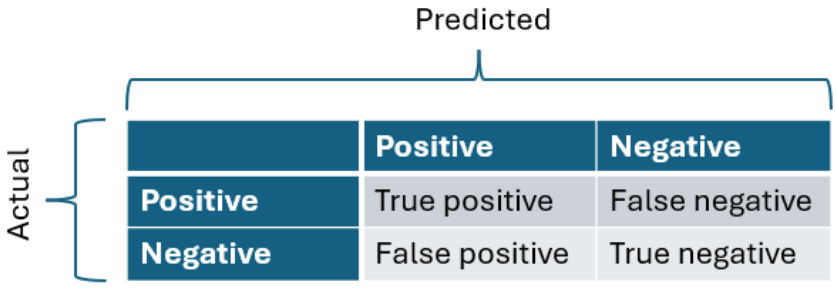


Figure 3.6: Example of the terms true positive, false positive, false negative, and true negative in a confusion matrix.

This category contains instances that were labeled as part of the positive class by the model, but are actually part of the negative class. Thirdly, the opposite of a false positive is the category of *false negatives*. This category contains instances that are labeled as negative by the model, but are actually positive. Finally, the category of *true negatives* contains all samples that are correctly labeled as negative.

Precision reflects how many out of the full set of labels that were predicted to be of a certain class, are correctly labeled. In other words, precision is the ratio between the true positives and the full group of true positives and false positives. Recall is used to reflect how many of the labels that are actually of a certain class are predicted to be so by the model. This is the ratio between the true positives and the group of true positives and false negatives combined. The F1 score is the harmonic mean of the precision and recall. In this thesis, I used the macro F1 score, which means that I first calculate that F1 score for each of the classes and then take the unweighted average of those scores. This means, that each class is equally important, independent of sample size.

The conditions differ based on what part of the data is used to train and test the model. For an overview of what is included in each of the conditions, see Figure 3.4 For the positive class, I used a total of six conditions: PUOH-only, OHD-only, HT+TL-only, PUOH+OHD, OHD+HT+TL, and All-Variants. I merged the thumb lift and the hand turnout subclasses because there were only 12 instances of a thumb lift gesture in the data set and this is not sufficient to train a model on, or to draw any generalizable conclusions based on the results. Comparing the PUOH-only, OHD-only, and HT+TL-only conditions will reveal if some of the gesture variants are easier to recognize than others due to, for example, the size of the gesture or the amount of data available. Comparing the models' performance on the other conditions to the performance on the one-subclass-only conditions will reveal to what extent including extra variation, as is the case in real life, will affect the models' performance. It should be noted that by combining these categories the size of the training and test data also increases, which could also lead to better performance.

For the negative class, I only used two conditions: with-US and without-US. This comparison will test how well the models are able to distinguish between the two classes when including the *unsure* subclass, which looks like palm up open hand gestures but might not always be intended that way by the speaker. While the real life data I collected includes these *unsure* gestures, they would be hard to annotate, even by a human annotator. Therefore, I tested the effects of including them in the data by

making a conditions with and a condition without them.

For the dynamic model I was not able to test all conditions due to the fact that the model took longer to run, even with a limited number of epochs. Therefore I decided to keep both conditions for the negative class, but limit the positive class to only the OHD-only condition and the All-Variants condition to compare maximum difference in the amount of variation within the positive class while keeping the number of positive instances in the data set as high as possible.

	Unsure	HT+TL	OHD	PUOH
HT+TL-only/With-US	✓	✓		
HT+TL-only/Without-US		✓		
OHD-only/With-US	✓		✓	
OHD-only/Without-US			✓	
PUOH-only/With-US	✓			✓
PUOH-only/Without-US				✓
HT+TL+OHD/With-US	✓	✓	✓	
HT+TL+OHD/Without-US		✓		
OHD+PUOH/With-US	✓		✓	✓
OHD+PUOH/Without-US			✓	✓
All-Variants/With-US	✓	✓	✓	✓
All-Variants/Without-US		✓	✓	✓

Table 3.4: Overview of what parts of the data are included in each of the twelve conditions used to test the static model with the conditions that are also used to test the dynamic model in bold.

3.4.1 Follow-up Experiment

One of the downsides of the experiment as described above is that the limiting the variation in types of palm up open hand gestures also limits the number of instances in the positive class. To determine to which extent this affects the models' performances, I performed a small follow-up experiment. Because of the long runtime of the dynamic model, I only performed this experiment for the static model.

To be able to compare the performances of the models in different conditions without sample size as confounding variable, I ran the same experiment again with a sample size of 250 images in both classes. For the negative class, I kept both the Without-US and the With-US conditions. However, for the positive class, I only compared the All-Variants condition to the OHD-only condition in order to compare the performance of the model with maximal variation in the positive class to the performance with minimal variation.

Chapter 4

Results

In this chapter, I present the results of the experiments I described in the previous chapter. I will do so by comparing the accuracy, precision, recall, and F1 scores of each of the different conditions I tested the models in. Furthermore, I provide confusion matrices and discuss which classes were easy to detect and which classes were commonly confused.

4.1 Static Model

4.1.1 Main Experiment

Table 4.1 shows the accuracy, macro F1 score, precision, and recall of the static model across different conditions. The highest recall and F1 score can be found in the All-Variants/Without-US condition: the model yielded a recall of 0.60 and an F1 score of 0.75. The highest accuracy was 0.91 and was found in both the With-US and Without-US variants of the HT+TL-only conditions and in the PUOH-only/Without-US condition. The highest precision was 0.83 and was achieved in the HT+TL-only/Without-US condition.

The lowest precision, recall, and F1 score were 0.33, 0.03, and 0.50 respectively and were yielded by the model trained in the HT+TL-only/With-US condition. The lowest accuracy was 0.79 and was achieved by the All-Variants/Without-US model.

Of the three conditions that only consider one gesture variant of the palm up open hand gesture, the overall best results were found in the OHD-only condition, that outperformed the other two one variant-only subclasses in both the With-US and Without-US variants. In the Without-US condition, the model achieved a precision of 0.75, a recall of 0.45, an F1 score of 0.74, and an accuracy of 0.85.

Overall, the Without-US conditions yielded better results than their With-US counterparts. Precision, recall, F1 score, and accuracy were better in almost all cases. Apart from the above-mentioned accuracy in the HT+TL-only conditions, which was equal in the With-US and Without-US conditions, the PUOH+OHD condition is the only exception. In this condition, the With-US model achieved a slightly higher precision than the Without-US model (0.74 compared to 0.73), as well as a slightly higher accuracy (0.82 compared to 0.81).

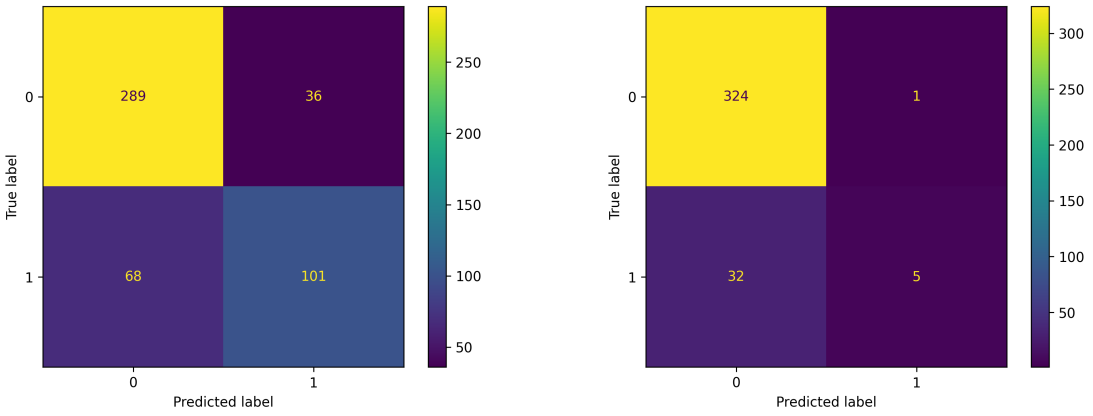
Figure 4.1 shows the confusion matrices of the All-Variants-only/Without-US and the HT+TL-only/Without-US condition. The confusion matrices for the other conditions can be found in Appendix C. In the All-Variants/Without-US condition, the

	With-US				Without-US			
	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc
PUOH-only	0.60	0.21	0.63	0.90	0.71	0.35	0.71	**0.91**
OHD-only	0.65	0.35	0.68	0.84	0.75	0.45	0.74	0.85
HT+TL-only	0.33	0.03	0.50	**0.91**	**0.83**	0.14	0.59	**0.91**
PUOH+OHD	**0.74**	0.49	**0.74**	0.82	0.73	0.55	**0.75**	0.81
OHD+HT+TL	0.68	0.41	0.69	0.80	0.70	0.55	0.74	0.81
All-Variants	0.71	**0.53**	0.73	0.78	0.74	**0.60**	**0.75**	0.79

Table 4.1: Precision, recall, macro F1 score and accuracy of the static model.

true negatives are the most common with 290 out of a total 494 instances. The second largest class is that of the true positives with 108 instances followed by the false negatives with 61 instances. The smallest category is that of the false positives with 35 instances.

In the HT+TL-only/Without-US condition, the largest category is also the true negative category with a total of 316 out of 372 instances. The second largest class is the false negatives with 30 instances. There were only 9 false positives and 7 true positives, indicating that the model only predicted a positive label for a total of 16 instances.



(a) All-Variants/Without-US

(b) HT+TL/Without-US

Figure 4.1: Confusion matrices for the static model.

4.1.2 Follow-Up Experiment

The results of the follow-up experiment are shown in Table 4.2. When keeping the sample size consistent across conditions, the results become a lot more similar, indicating that a lot of the variation that was visible in the initial results might be a result of class imbalance. While the OHD-only/With-US, the OHD-only/Without-US and the All-Variants/Without-US conditions all yield an accuracy and F1 score of 0.76, the All-Variants/With-US model performed worse than the other three conditions with an F1 score and accuracy of 0.70.

When comparing these results to the results of the main experiment, we can see that the model improved in terms of precision, recall, and F1 score in nearly all conditions.

	With-US				Without-US			
	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc
OHD-only	**0.76**	0.76	**0.76**	**0.76**	0.74	0.75	**0.76**	**0.76**
All-Variants	0.70	0.71	0.70	0.70	0.75	**0.77**	**0.76**	**0.76**

Table 4.2: Precision, recall, macro F1 score and accuracy of the static model with a set sample size of 250 images per class.

The accuracy of the model decreased in the follow-up experiment. This is likely due to the fact that accuracy, as explained in the methodology section, is highly sensitive to class imbalance. Because of this, it will yield higher scores if the model is only good at predicting the majority class, as was the case in the main experiment.

4.2 Dynamic Model

Table 4.3 shows the macro F1 score, precision, recall, and accuracy of the dynamic model across the four conditions it was trained and tested in. In all four metrics, the best performance is found in the OHD-only/Without-US condition with a precision, recall and F1 score of 0.53 and an accuracy of 0.59. The condition with the worst recall, F1 score, and accuracy is the opposite condition: the All-Variants/With-US condition. There, the model yielded a recall of 0.34, an F1 score of 0.26 and an accuracy of 0.30. The precision was significantly higher with a score of 0.48. The lowest precision was 0.35 and was found in the All-Variants/Without-US condition.

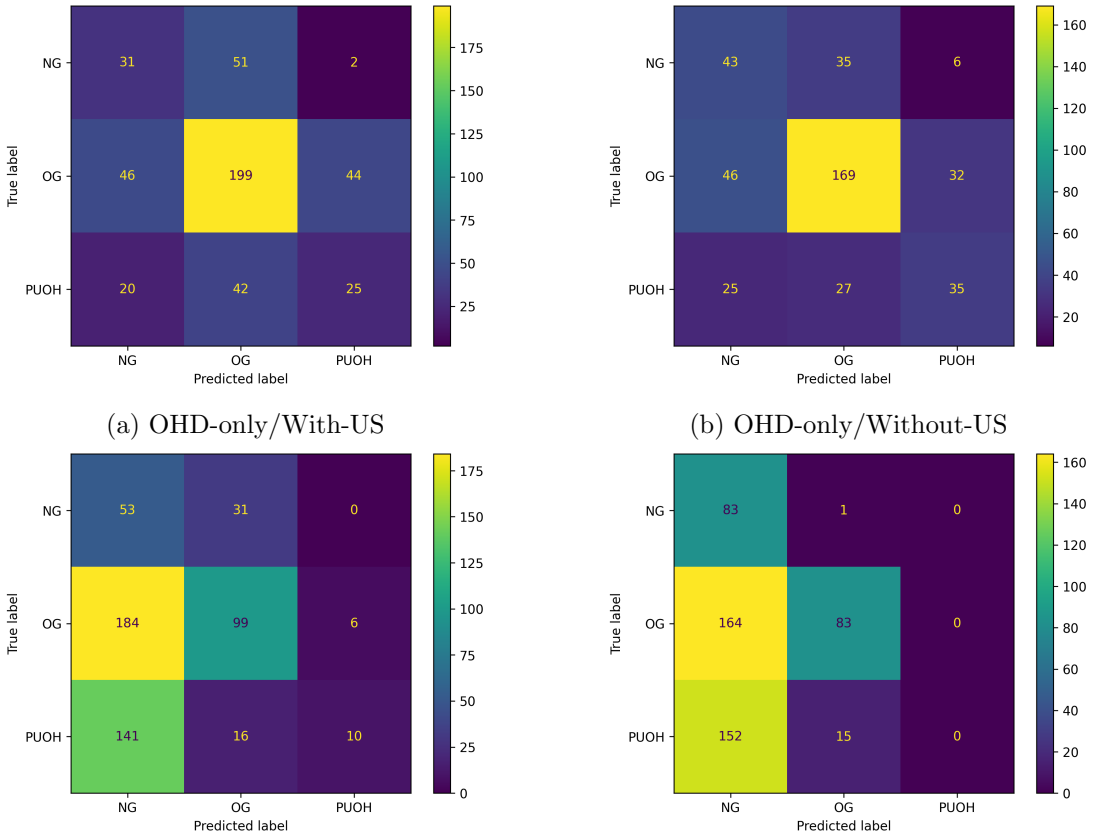
	With-US				Without-US			
	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc
OHD-only	0.45	0.45	0.45	0.55	**0.53**	**0.53**	**0.53**	**0.59**
All-Variants	0.48	0.34	0.26	0.30	0.35	0.45	0.27	0.33

Table 4.3: Precision, recall, macro F1 score and accuracy of the dynamic model.

Figure 4.2 shows the confusion matrices for each of the four conditions. We can see that the negative gesture class is the most commonly predicted for both OHD-only conditions while for the PUOH+OH+HT+TL conditions, the No gesture class in the most common. In the OHD-only/With-US condition, 199 out of a total of 292 negative gesture videos were classified correctly. In the OHD-only/Without-US condition this was the case for 169 out of 231 instances. In the OHD-only, the negative label was also most commonly predicted for both other classes: in the no gesture class, which has a total of 84 instances, 51 were incorrectly labeled as negative, compared to only 31 correctly labeled instances. Out of the total of 87 positive instances, 42 were incorrectly labeled as negative compared to 25 correctly labeled ones. In the OHD-only/Without-US conditions, the most common label was the correct label for both the positive and the no gesture classes. Out of the 87 positive gestures, 35 were correctly labeled as positive, compared to 25 labeled as No gesture and 27 labeled as negative. In the no gesture class, which had a total of 84 instances, 43 were labeled as no gesture, compared to 35 as negative and 6 as positive.

In the All-Variants conditions, the most common predicted label for all classes was no gesture. In the All-Variants/With-US condition, this label was predicted for 378 out

of 540 videos, only 53 of which were correct. The second most common label was the negative label, which was predicted 146 times, 99 of which were correct. The positive label was only predicted 16 times, 10 of which were correct, despite 167 positive gestures being present in the test set. In the All-Variants/Without-US condition, the positive label was never predicted, even though there was a total of 167 instances in the test set. The most commonly predicted label was the no gesture label. Of the total of 498 videos, this label was predicted 399 times, 83 of which correctly.



(a) OHD-only/With-US

(b) OHD-only/Without-US

(c) All-Variants/With-US

(d) All-Variants/Without-US

Figure 4.2: Confusion matrices for the dynamic model.

4.3 Participant Analysis

To study the effect of individual variation on the models' performances, I calculated the macro F1 score per participant for the best performing static model and the best performing dynamic model from the main experiment. These were the All-Variants/Without-US static model and the OHD-only/Without-US dynamic model. The results are shown in Table 4.4.

Both models show a large variety in participant F1 scores. For the static model, the scores fall within a range of 0.38 until 1.00. For the dynamic model, this range is slightly smaller, with every score falling between 0.32 and 0.84. It is interesting to note that the participant with the highest F1 score in the dynamic model, participant 2L, has the lowest F1 score in the static model. For both models, this participant only

has a small number of instances in the test data (5 instances in the static model and 7 instances in the dynamic model). For the static model, there seems to be a positive correlation between sample size and F1 score. To assess the size of this correlation, I computed Pearson's correlation coefficient. I found a significant positive correlation (r=0.46, p=0.04). One exception to this pattern stands out: participant 2R, whose gestures yielded a perfect F1 score of 1.00, only has a sample size of 4. However, due to the very small sample size, this could be due to chance. For the dynamic model, I found an insignificant weak negative correlation (r=-0.28, p-0.23), meaning that the model performs slightly better for participants with a smaller sample size, but not significantly so.

	Static		Dynamic	
Participant	F1	Sample size	F1	Sample size
1L	0.44	5	0.44	5
1R	0.71	38	0.52	27
2L	**0.38**	5	**0.84**	7
2R	**1.00**	4	0.46	9
3L	0.60	41	0.34	45
3R	0.75	33	**0.32**	19
4L	0.46	13	0.49	13
4R	0.42	11	0.76	14
5L	0.63	28	0.54	28
5R	0.42	18	0.66	15
6L	0.69	23	0.35	20
6R	0.79	40	0.57	31
7L	**0.85**	61	0.50	45
7R	0.75	24	0.50	16
8L	0.80	15	0.43	9
8R	0.81	34	0.58	32
9L	0.60	30	0.44	26
9R	0.80	33	**0.32**	34
10L	0.78	24	0.33	12
10R	0.36	14	0.61	10

Table 4.4: Macro F1 score and sample size in test set per participant for the All-Variants/Without-US condition of the static model and the OHD-only/Without-US condition of the dynamic model in the main experiment.

4.4 Main Takeaways

For the static model, the condition in which the model performed best was the All-Variants/Without-US condition which achieved the highest F1 score and recall. In almost all conditions, the With-US condition yielded worse results than the Without-US condition, indicating that excluding the *unsure* category improves model performance. Overall, the model performed better in cases where a larger part of the positive class instances were included.

For the dynamic model, the OHD-only/Without-US condition yielded the best results in all four metrics. Just as was the case for the static model, excluding the

unsure subclass improved the results in all conditions. However, while the static model performed best with more positive cases in the data set, the opposite is true for the dynamic model. Here, the OHD-only conditions yield better results than the All-Variants conditions, indicating a preference for less variation in form within the positive class.

When looking at the variation per participant, one can see that there is a large amount of individual variation present in both models. In the static model, there is a positive correlation between model performance and number of gestures by that participant in the data set. There is no such correlation present for the results of the dynamic model.

Chapter 5

Discussion

The discussion consists of three parts. First, I provide a more detailed interpretation of the results, discussing possible explanations and implications. Then I describe an error analysis I performed for both models. Finally, I compare the two models and discuss some limitations of this thesis.

5.1 Static Model

5.1.1 Interpretation of the Results

Main Experiment

Out of all of the static models, the static model in the All-Variants/Without-US condition seems the most promising, as the recall and F1 score are the highest in this condition and higher precision scores are only found by models with significantly lower recall scores. The results of this model are better than chance, and while there is still much improvement to be made, these results indicate that static recognition of palm up open hand gestures is possible and further research into improving these models would be worthwhile.

The results of the static model strongly correlate with the relative size of the positive class in each of the conditions. The conditions with the highest recall, precision, and macro F1 score are the conditions with the most positive instances such as the All-Variants/Without-US condition. However, for accuracy, the opposite seems to be true: the conditions with fewer positive gestures, such as the HT+TL-only conditions and the PUOH-only conditions, yield the highest accuracy. This can be explained by the fact that accuracy as a metric is much more sensitive to class imbalance and always predicting the majority class can lead to a higher accuracy.

This is further supported by the confusion matrices. Especially in the HT+TL/Without-US condition, where false negatives were more common than true and false positives combined, the model predicted the negative majority class in almost all instances. In the All-Variants/Without-US condition, the results seem more balanced. The second most common category is the true positive category, indicating that the model does not only predict the majority class in this condition.

The per participant analysis also shows that this model is very sensitive to sample size. The gestures made by participants with a larger number of gestures in the data set were easier to classify correctly. This confirms that there is variation between individuals in the form of palm up open hand gestures. The model has learned to

recognize palm up open hand gestures based on the features that present in the majority of the data, making it more difficult to recognize palm up open hand gestures for participants with a smaller sample size.

Follow-Up Experiment

The follow-up experiment showed that when keeping sample size consistent across conditions, the static model performs worst in the All-Variants/With-US condition. This can be explained by the fact that this is the condition with the greatest variation within both the negative class and the positive class. Features that play an important role in recognizing palm up open hand gestures in one of the subclasses might not be an important indicator of a palm up open hand gesture in other subclasses, making it harder for the model to generalize. Furthermore, the addition of the *US* subclass to the Other class can make this more complicated by adding gestures that are very similar to palm up open hand gestures in form.

Based on the premise that increased variation within a gesture makes it harder for a gesture recognition model to recognize these gestures, one would expect to see a difference between the OHD-only/Without-US condition and the two remaining conditions. In this condition, the positive class fully consists of instances taken from the *OHD* subclass and the US subclass is excluded. However, the results of this condition are not better than those of the other two, indicating that it is the combination of variation in the positive class and gestures that are similar to positive gestures in the negative class that makes it difficult for the model to accurately classify the gestures in the All-Variants/With-US condition. This could be a result of two things. Firstly, it could be the case that the *unsure* instances in the sample look similar to either *HT+TL* or *PUOH* gestures. The All-Variants/With-US condition is the only condition where all three types of gestures are present and therefore the only condition where the model could confuse them. The second option is that the model has a broader idea of what a palm up open hand gesture is due to the larger amount of variation. Even if the *unsure* gestures do not look like *HT+TL* or *PUOH* gestures, the increased variation could cause the conditions needed for classifying a gesture as positive to become less specific. As a result of this, it could be the case that the *unsure* gestures, which already look similar to palm up open hand gestures, now fall within those conditions.

When comparing the results of the follow-up experiment to the main experiment, I found that the model improved in the follow-up experiment in terms of precision, recall, and F1 score in almost all conditions. This indicates that the model was negatively affected by the difference in sample size in the main experiment. Furthermore, it shows that including more data does not always lead to better results, especially when including more data results in class imbalance.

5.1.2 Error Analysis

I performed two error analyses to gain more insight in the performance of the static model. In the first error analysis, I took a sample of 10 images from each of the sets of images classified as true positives, true negatives, false negatives, and false positives. I then looked at each of the images and the videos they were taken from to see if this would give more insight in why the model classifies these images (in)correctly and to see if patterns would become apparent. I did this analysis both for the All-Variants/Without-US model, which had the best performance, and the HT+TL-only/Without-US model,

which performed worse. In cases where there were not enough instances available for me to take a sample of 10 instances, I have taken as many as possible and mentioned this in the discussion.

The second error analysis I carried out is based on the same method, but instead of taking samples of the different categories of the confusion matrix, I took 10 samples based on the subclasses. In order to study the full range of subclasses, I chose to use this analysis for the All-Variants/With-US condition.

True Positives

For the All-Variants condition, my sample contained seven open hand diagonals, two hand turnouts and one full palm up open hand gesture. All images were very easy to recognize as a palm up open hand gesture, even without looking at the full video. In two images, the fingers of the hand were not entirely visible because of the way the side of the video was trimmed off to remove the other participant from view. In these cases, this did not affect the model's performance. This indicates that the model is able to estimate the positions of the missing joints well enough to not negatively affect performance.

In the HT+TL-only condition, there were only five true positives, four of which were hand turnouts and one was a thumb lift. Although all five were recognizable as part of the positive class by just looking at the image, some of the images were made from an angle that makes it difficult to see exactly how the hand is being held due to part of the hand being occluded. An example of this is shown in Figure 5.1. In the images sampled here, the model was still able to correctly detect the hand and classify the gesture.



Figure 5.1: Example of a PUOH gesture where part of the hand is occluded. This image is mirrored, the gesture is made with the right hand.

False Positives

In the All-Variants condition, the false positives sample contained four no gesture images and six other gesture images. Although several of the images in this category are participants with their hands intertwined in their lap, three images show a participant with their hand open, just not facing up. An example of this can be seen in Figure 5.2. This image shows the participant with her hand fully open, but facing downwards. This indicates that the model might be able to recognize the shape of the hand but struggles with the orientation. There are also two images that look like palm up open hand gestures when only considering the image. However, when looking at the video, it becomes clear that these gestures are not a palm up open hand gesture but a part of another gesture. An example of this can be seen in Figure 5.3 that looks like the hold of a palm up open hand gesture but is instead part of a cyclic gesture towards the speaker. In these cases, there is no way for the model to determine that these gestures are not palm up open hand gestures with the input is it is given. This could be solved by, for instance, combining it with a dynamic model, or by including several images as input for each gesture.

The sample for the HT+TL-only condition only contains one instance of an other gesture subclass labeled as positive.



Figure 5.2: Example of a participant with an open hand facing down. This image is mirrored, the speaker is making the gesture with her the left hand.

True Negatives

The sample for the All-Variants condition contains one self-adapter, one no gesture, and eight other gestures. Although most images clearly do not contain a PUOH gesture, some instances are very similar to a palm up open hand gesture but are still correctly classified as negative. An example of this can be seen in Figure 5.4 where the



Figure 5.3: Example of a gesture in the negative class that looks like a palm up open hand gesture. This is visible in both hands.

participant's left hand seems to be in a palm up open hand gesture. However, when viewing the full video, one can see that while the participant is leaving her left hand facing palm up, the gesture the participant makes is made with her right hand. As I discussed before, the model only uses input features from one hand. Therefore, it is likely that the model took the input features from the hand hand that was making the other gesture instead of from the hand that was facing palm up. In this case, this prevented the model from incorrectly classifying an image as positive. However, in cases where a positive gesture is only made with one hand, it could lead the model to wrongly classify this as negative because it is using the hand in resting position to compute the features.

In the HT+TL condition, there are two no gesture images, three self-adapter images and five other gesture images in the true negatives sample. All of the images are clearly not part of a palm up open hand gesture. However, it is worth noting that despite none of the images containing hands that were both palm up and open, there are several open hands present in this sample. While this caused problems in the All-Variants condition, as was discussed in the false positives section, the model is able to correctly classify these images in the HT+TL condition. This might be due to the fact that in hand turnouts and thumb lifts, the hand is generally not fully open. Because of this, it is likely that the model did not use features based on the extension of the fingers to determine to which class an image belonged.

False Negatives

The All-Variants false negatives sample contains three palm up open hand gestures, five open hand diagonals, one hand turnout, and one thumb lift. While all of these images clearly show the hold of a palm up open hand gesture, in some images part of the hand(s) is occluded or out of focus, as can be seen in Figure 5.5. This might have



Figure 5.4: Example of a correctly classified negative gesture that looks similar to a palm up open hand gesture. The gesture is made by the left hand.

led to mistakes in determining the landmark positions for these images. Furthermore, for false negative gestures, it is once again very important to note that the model only bases the landmark features off of one of the hands. Therefore, in cases where a gesture is made using only one hand, there is no way of knowing whether the model wrongly classified the hand making the palm up open hand gesture, or if the model simply looked at the wrong hand.



Figure 5.5: Example of a palm up open hand gesture where the right hand is out of focus.

In the HT+TL-only condition there was one thumb lift image and nine hand turnout images in in the sample for false negatives. These gestures are smaller, and less noticeable than was the case in the All-Variants condition. They look more similar to self-adapters and in some cases, such as is shown in Figure 5.6, they are even combined with self-adapter gestures. In these cases, the model could have mistakenly classified the self-adapter gesture instead of the palm up open hand gesture.



Figure 5.6: Example of a hand turnout that is happens simultaneous with a self-adapter.

Positive class

The second error analysis was based on the eight subclasses. There were only five thumb lifts present in the test set, two of which were correctly classified as positive gestures. Although the hands are not always fully visible due to the participants resting their hands in their lap, all five of the images clearly show the thumb lift. There seem to be no clear differences between the correctly and incorrectly classified instances.

In the sample of hand turnouts, there were five true positives and five false negatives. Among the false negatives, there are two instances of hand turnouts where the participant is touching their hair with their other hand. Because touching one's hair is a self-adapter, it could be the case that the model classified these images as negative because of similarities with self-adapter images. It is worth noting that in one of the correctly classified images the participant is sitting sideways on the chair, hiding her arm behind her upper body as is shown in Figure 5.7. Because the static model only uses the landmarks of the hand, its ability to recognize the gesture was not impacted by the occlusion of the arm and shoulder.

The open hand diagonal sample was also equally divided across true positives and false negatives. Four out of the five correctly classified open hand diagonals show a



Figure 5.7: Mirrored image of a participant with her right arm hidden behind her body.

posture similar to the one shown in Figure 5.8, where one hand is placed in the lap of the participant and the other hand is making a clear open hand diagonal gesture somewhere in the air. In the incorrectly classified instances, it is more common for participants to have their hands facing the camera, such as is the case in Figure 5.9. This indicates that the position of the hand might be influential to the model's performance. This can be explained by the fact that the landmarks that are used as input for the model depend on the position of the hands relative to the camera.

Of the images in the palm up open hand gesture sample, six were correctly recognized and four were incorrectly classified as negative gestures. There is a mix of different positions and camera angles in both the correct and incorrect images, making it difficult to draw conclusions about the functioning of the model based on this subset.

Negative class

The sample from the no gesture subclass only contained images that were correctly classified as negative. It is expected for the model to perform relatively well on this subclass because the no gesture subclass mainly consists of pictures of participants having their hands resting in their lap.

The other gesture sample consists of nine true negatives and one false positive. While the true negatives contain a mix of different hand positions, in eight of them the palms of the hand of the participants are facing down. The participant in the false positive image, shown in Figure 5.10 does have both his palms facing upwards, but does not extend his fingers as would be the case in a positive gesture.

The self-adapter sample only consists of true negative instances. In nine of out ten images, the participant is in some way touching their face or hair. As a result of this, the hands are located in a different part of the image then is usually the case in a



Figure 5.8: Mirrored image of a participant with her left hand in her lap and her right hand in the air while making a palm up open hand gesture.



Figure 5.9: Example of a participant with his hands making a palm up open hand gesture towards the camera.

positive gesture. On top of this, the palms are often facing away from the camera and the hands are often either closed or only partially open. An example of this can be seen in Figure 5.11. This is very different from the palm up open hand gesture, which is a convincing reason for the fact that the model was able to perform well on classifying this subclass.

The sample that I took for the unsure category only contained one false positive and nine true negatives. This was unexpected, because the form of the unsure gestures was often more similar to the form of the palm up open hand than any other gesture

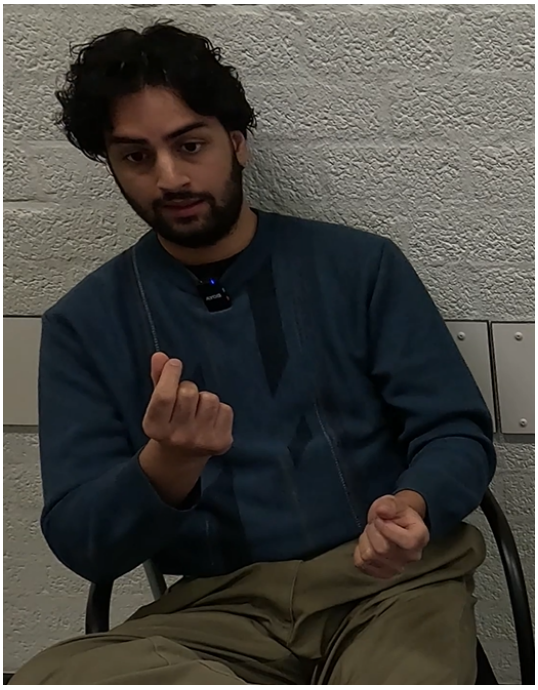


Figure 5.10: Example of a participant with his palms facing upwards and his hands closed.



Figure 5.11: Mirrored example of a self-adapter image where the left hand is partially open and the palm is facing away from the camera.

in the unsure category. The one false positive is shown in Figure 5.12. Just as was the case for one of the false positives described above, the gesture in this image does show the characteristics of a palm up open hand gesture. However, when viewing the

video, we can see that the movement is being made towards the participant instead of away from them. The participants in the nine true negative images all have their hands partially or fully open with their palms either diagonal or facing up in all but one image. Without looking at the videos these gestures are part of, one would probably classify these images as positive class gestures. However, the model was able to correctly classify these images, possibly based on features that stand out less to human annotators. It could also be the case that there are simply more images classified as the negative minority class and that this is the reason instances are more likely to be classified as negative. More research is needed to study if class imbalance is indeed the reason for the model's unexpected performance on the unsure category or if there is some difference between the positions of the hands in the unsure subclass and the positive class.



Figure 5.12: Image of a participant making a gesture towards her upper body that looks similar to a palm up open hand gesture. The gesture is made with both hands.

Summary

More research is needed to draw conclusions about the performance of the static model with certainty due to the limited sample size, and further research is needed to test if these findings are true on a larger scale. However, this error analysis can still point us in the direction of potential limitations of the model that could be improved upon in further research.

The first error analysis had two main takeaways. Firstly, the effect of (part of) the hand being obscured is still unclear. While there were several images where the hand was out of focus, out of frame, or hidden behind another body part or object in the false negative sample set, there were also some instances where the image was classified correctly despite the tips of the fingers not being fully visible.

Second, in the full All-Variants model, images in which the hand was open but not palm up were often wrongly classified as a positive palm up open hand gesture, while this was not the case in the HT+TL model. This could point to the model being able to correctly capture the shape of the hand but struggling with the orientation.

The second analysis also has two main findings. The first is that, in addition to

the rotation of the hand possibly causing problems with detecting palm up open hand gestures, the position of the hand relative to the camera might also increase the difficulty of this process. There were several images where the exact position of the hand and fingers were difficult to see in the image due to the participant gesturing towards the camera instead of to the other participant.

Secondly, the group that was predicted as the negative class contained several true negatives that a human annotator would most likely wrongly classify as positive if they were unable to watch the whole video. While more research is needed to see if there are any differences in hand position that did not stand out to me, this could also be highly influenced by the negative class being the majority class in the training data.

In both analyses, I found false negatives where the participant made the palm up open hand gesture with only one of their hands. These images could be classified incorrectly because the models only take the landmarks of one hand as input. If the palm up open hand gesture was only made with one hand, it could have been the case that the model simply used the landmarks of the hand that is not making the palm up open hand gesture.

5.2 Dynamic model

5.2.1 Interpretation of the Results

The condition that yielded the best performing model is the OHD-only/With-US condition, followed by the OHD-only/With-US condition. The two All-Variants models yielded considerably worse results in all metrics except for precision. Even though there is a larger class imbalance in the OHD-only condition, the decrease in variety in the positive class leads to better overall results.

When inspecting the confusion matrices, we can see that the models trained in the All-Variants conditions rarely predict the positive label. This is unexpected given the fact that over 30% of the instances in the training data were of the positive class. It is likely that the high amount of variation caused the model to always have some level of uncertainty, therefore not allowing the positive class probability to go over the 0.9 threshold.

The per participant analysis shows that there is a large amount of variation per participant. No significant correlation between per participant model performance and participant sample size was found. However, while there is no way of knowing why the model performs better for certain participants without additional research, it is clear that this model, like the static model, is sensitive to individual variation.

5.2.2 Error Analysis

In the error analysis I performed for the dynamic model, I focused on the model with the best performance, which was the model in the OHD-only/Without-US condition. I took a random sample of five videos for each of the categories shown in the confusion matrices and studied them in more detail to gain a better understanding of the model's performance.

True Label: No Gesture

The no gesture category was annotated by selecting random parts of the video that did not receive any other annotation label. Because, as explained above, I did not label all self-adapter gestures for their full duration, it can be the case that there are some minor self-adapters present in this category even though the self-adapter subclass falls under the negative class.

In the sample of videos that were correctly labeled No Gesture, there were two videos where there was no movement at all. In two videos there was a small movement of the fingers, which might be part of a very small self-adapter gesture. One video showed a participant moving his upper body backwards in order to sit straight up in his chair. While he is doing so, his arms are crossed and his hands are not visible to the camera.

In the five videos that have the true label no gesture but received the negative label, there were slight movements of the hand that could have been seen as self-adapters. One video showed a larger movement of a participant moving his hands back together after seemingly having grasped a nonexistent object in the air by having two open hands facing each other. Without the context of the full video it is hard to say whether this is the retraction of a previous gesture of a separate gesture by itself. However, this video should have been labeled negative and is therefore a result of a mistake in the annotation process.

In the sample of No Gesture videos that were labeled positive by the model, there are three videos that contain small movements of the fingers. During one video, the participant has his open hand on his lap with his palm facing upwards. While the rotation movement that is part of a palm up open hand gesture was not visible, this resting position could have easily been confused with the hold of a palm up open hand gesture. In one video, the hands of the participant are not visible because his arms are crossed.

True label: Negative

In the sample of gestures that were correctly labeled as negative, there were two self-adapters and three other gestures. One of these three gestures includes a rotation of the hand and lower arm but is not made with an open hand or an upwards facing palm.

Given the fact that, as I described above, it is likely that there were several smaller self-adapter gestures present in the no gesture training data, I expected most of no gesture-negative errors to be self-adapters. However, in the sample of videos that were labeled no gesture but had the true label negative, there was one self-adapter present. There were also two very small other gestures and one bigger one where the participant moved his closed hands towards his chest. Finally, there was one video where something has gone wrong in processing the video clip, which led to this clip being a frozen image. For a frozen video no gesture would have been the correct label, so this error cannot be attributed to bad performance by the model.

The five videos in which the positive label was assigned instead of the true label negative all show participants with an open hand with the palm facing downwards. This strongly suggests that while the model is able to pick up on the hand being open, the orientation of the hand does not play a role in assigning a label to a gesture.

True label: Positive

The five videos that are correctly labeled positive all show a clear but not necessarily very large palm up open hand gesture. It is worth noting that in one of the videos the hand making the palm up open hand gesture briefly moves out of the frame. However, this did not impact the model's performance to classify this gesture.

In the sample of positive videos that received the label no gesture, four out of the five videos show a very clear palm up open hand gesture where the retraction almost immediately follows the stroke. If it is indeed the case that the model recognizes a palm up open hand gesture by the position of the hand during the hold, having a very short hold might negatively impact the models' ability to recognize this position.

The sample of positive videos labeled as negative contains two very clear palm up open hand gestures with the hand facing up and the fingers fully extended. In two of the other videos, there is less extension of the fingers. Instead the participants keep their hand in a more relaxed position. In the final video, the participant does fully open his hand but only slightly turns his wrist to have his palms facing upwards.

Summary

There are three main takeaways from the error analysis above. First, because the No Gesture clips are randomly extracted from parts of the video that received no annotation during the annotation process and self-adapters were not always labeled, there might be some cases where a No Gesture gesture actually is a self-adapter. For the static model, these subclasses were merged in training and initial testing. However, for the dynamic model, these categories are separate. Because any gesture first needs to pass the motion threshold, distinguishing between the No Gesture class and the two other classes, irregularities in annotations for this class could have affected the performance of the model during the recognition of palm up open hand gestures.

Second, just as was the case for the static model, the dynamic model seems to struggle with recognizing the rotation of the hand. There were multiple videos incorrectly classified as positive while the open hand was facing down instead of up. This could be an indication that the model uses the openness of the hand as main feature to classify gestures as positive.

Finally, there were several videos where the true label was positive that were incorrectly labeled, potentially because the retraction followed the stroke almost immediately, shortening the duration of the hold. While further research is needed to test which part of the palm up open hand gesture is recognized by the model, the findings above indicate that the fully open hand, which takes place in the hold phase, plays a larger part in this than the opening or rotation of the hand.

5.3 Comparison

In this section, I compare the requirements and performances of the two models. An overview in shown in Table 5.1. Overall, the static model performed better than the dynamic model in both the main experiment and the follow-up experiment. Not only did it score higher in the evaluation metrics, but it also took under a minute to train a model and predict the labels compared to the 10 hours it took for training the dynamic model and classifying the test videos.

However, there is more manual labor involved in the static model. While the dynamic model could directly use the video clips extracted from the original data based on the annotations, the static model required taking screenshots of the hold of each gesture manually. On the other hand, this manual selection did ensure that the model was training on the same part of the gesture in every instance. In the dynamic model, the gesture occurs somewhere in the video but it is not noted what phase of the gesture takes place at what point in the video. Therefore, the model not only needs to correctly classify the gesture, but also recognize that the gesture is happening, leading to bad performances on palm up open hand gestures with a short hold.

While the dynamic model performed better in the OHD-only condition, which has less in-class variation but a larger class imbalance, the static model performed better in the All-Variants condition, for which the opposite is true. For natural data, a large class imbalance and a lot of variation in a set of gestures is very common, and therefore the static model would most likely be the best choice. However, in more artificial settings, the dynamic model might better suit the data. A possible explanation for the fact that the static model is less affected by the increased amounts of variation in form than the dynamic model is, is that the static model only uses features based on the hand shape as input, while the dynamic model uses input features based on the full body. Therefore, movements of the arm or shoulder, which are common in larger palm up open hand gestures, are ignored in the static model, but could be distracting in the dynamic model.

An advantage of the dynamic model is that it is able to take both hands, as well as the movement of the arms, shoulders, head, and upper body into account when classifying a gesture. In comparison, the static model only looks at one of the two hands, without stating which hand that is.

An advantage of the static model is that it is a binary model. Gestures are classified as either positive or negative, making classification as simple as possible. The dynamic model is more complicated, because it first classifies the data into gesture and no gesture, and then divides the gesture category into positive and negative. This could create more possibilities to adjust the model to classify different types of gestures or potentially classify variants of the same gesture. However, for this thesis, the addition of a third category is unnecessarily complicated.

For both models, the results are highly dependent on the angle of the camera and the relative position of the hand as well as the participant making the gesture. Furthermore, both models seem to focus on the openness of the hand and not on the direction the palm is facing. This can lead to the models making false negative predictions when the hand is open with the palm facing down.

5.4 Connection to the Literature

The fact that the models developed in this thesis are able to recognize palm up open hand gestures while there is a lot of variation in form present in the data, indicates that there are indeed features that are shared across all forms on this continuum, as was suggested by Cienki (2021). This is further supported by the error analyses I performed, which show that all gesture forms can be recognized correctly by the model at least some times, even if this is not always the case.

The features that are used by both models are most similar to the ones that are used to describe gestures in the Bressem et al. (2013) external system. However, the

	Static Model	Dynamic Model
Better overall performance	✓	
Requires little manual labor		✓
Performance is mostly affected by class imbalance	✓	
Performance is mostly affected by variation		✓
Input features are based on	hand	full body
Maximum number of hands in input	1	2
Number of predicted classes	2	3
Performance is dependent on camera angle	✓	✓
Performance is dependent on participant	✓	✓

Table 5.1: An overview of the differences and similarities in the requirements and performance of the static and dynamic models.

characteristic of a palm up open hand gesture the models' struggled recognizing the most is the rotation of the lower arm and supination of the hand. This is especially hard to capture based on the landmarks that are currently used in feature extraction. This led to the model often incorrectly classifying gestures were the participants opened their hand with the palm facing down as positive palm up open hand gestures. However, this is one of the key features used to describe palm up open hand gestures in the internal system developed by Boutet et al. (2018). This supports the idea proposed by Cienki (2021) that this system better captures the shared characteristics of all forms of palm up open hand gestures.

Additionally, this study supports Lutgen et al.'s idea (2026) that including variation in data sets, is essential to research in the field of NLP . Firstly, I found different forms on the continuum of palm up open hand gestures that was described by Cienki (2021) present in Dutch. The fact that all these forms exist and are used by speakers of Dutch indicates that focusing on one of them would not accurately capture the linguistic variety present in natural spoken Dutch.

Secondly, I have found differences in the frequency of palm up open hand gestures across participants. This supports the findings of Chu et al. (2014), who state that there is variation in gesture frequency among individuals.

Additionally, a reason to pay attention to the inclusion of individual variation in NLP data sets is the fact that the performance of the models showed a large amount of variation based on participant. For the static model, it was the case that participants with more samples in the data set had better performances. This means that, at least in this case, ensuring that there is a large variety of individuals represented in the data, and that their representation is equal, is not only important to accurately reflect real life language, but also to create the best performing models.

Lutgen et al. (2026) state that the choices in the data selection should be based on what should be represented in the data instead of by the amount. This is supported by the findings of my follow-up experiment in which I used a set sample size for the positive and negative class instead of using as much data as possible, which led to an improvement in the model's performance.

5.5 Limitations

In this section I discuss some limitations of the research done in this thesis. Firstly, I will discuss some limitations of the data.

One limitation is that all participants were university students. While this does not mean no conclusions can be drawn based on the data, it does require extra care when drawing conclusions. Because age and social group impact the way in which someone gestures are made (Feyereisen and De Lannoy 1991, Chu et al. 2014), it is not unlikely that another pool of participants, who perhaps differ in age would gesture differently. It could be the case that their palm up open hand gestures occur with a different frequency and have a slightly different form. To create a model that is able to recognize palm up open hand gestures when given any Dutch video data, it is important to have as much diversity as possible in the training data.

A second way in which my data is limited is that all videos captured the gestures from roughly the same angle. In this thesis, the models were only tested on videos that were made from the same angle as the training data. However, when creating a model with the aim of being able to classify any video or image containing a gesture, the model needs to be trained on a large variety of videos with different camera angles Nam and Wohn (1996). For these purposes, it is also important to test the models on a test set that differs from the training data in some way, to see if the model is able to generalize to new instances. However, due to my limited data set, this was not possible within this thesis.

There were also some limitations in the preprocessing and annotation of the data. Firstly, there was only one annotator annotating the entire data set. While it would have been preferable to have multiple annotators annotating parts of the data and calculating an inter-annotator agreement score, I was the only annotator available. This means that the quality of my annotations is not verified by another annotator. To ensure my annotations are reproducible, I included a detailed annotation manual with examples in Appendix B.

Secondly, during the error analysis I found that at least one of the videos was frozen. While I am unable to determine what caused this, it is most likely an error with FFmpeg, which I used to generate the video clips. It is possible that the same issue occurred in other videos too. This would have negatively impacted the model's performance.

Furthermore, there are two main limitations of the static model I used in this thesis. The first and most important limitation is that the model only used the landmarks of one of the hands of the participants as input. Because not all palm up open hand gestures are made with two hands, some of the false negatives yielded by this model could be due to the model using the features from the hand that is not making the gesture. Another limitation of the static model is the resolution of the video recordings. Many of the screenshots were blurred to the extent that the position and shape of the hand became less visible. This could have negatively affected Mediapipe's ability to recognize the landmarks in these images, which could have in turn impacted the model's ability to correctly classify the hand gestures.

A final limitation of this study is related to the training of the dynamic model. While the original website that the architecture of this model was posted on used 500 epochs with a patience of 10, I was only able to train the model with 5 epochs total. This is due to limited time and computational power. However, this could have

majorly impacted the results of the model, as the model was still improving when I cut the training time short. By increasing the number of epochs, the model might have continued to improve, which could have led to better performances.

Chapter 6

Conclusion

6.1 Summary

In this thesis, I originally aimed to answer the research question: "How can gesture detection models recognize recurrent gestures in Dutch?". Later, I narrowed this question down to: (1) "What forms does the palm up open hand recurrent gesture take in Dutch?" and (2) " How can gesture recognition models recognize these gestures?".

To investigate this, I created a small corpus of videos of two native speakers of Dutch having a naturalistic conversation. In this corpus I found that the palm up open hand gesture, which is known to be common in many spoken and signed languages (see for example Cooperrider et al. 2018), is also common in Dutch, with a total of 567 palm up open hand gestures in the full dataset. I annotated the video data to mark all instances of palm up open hand gestures as well as their form, which was adapted from Cienki (2021) and all other gestures, self-adapters, and some periods where there was no gesturing. Part of this corpus will be made available upon request, in accordance with the consent forms as signed by each of the participants. This corpus could in the future be used for further research on palm up open hand gestures, in which case the annotations would also be made available. However, this corpus could also be used for other types of linguistic research that focuses on natural conversation.

To test whether it was possible to recognize the palm up open hand gestures using gesture recognition models, I adjusted, trained, and compared two models. The first model was a static model, based on Tokhirov (2023), using images as input. The second model was a dynamic model, based on Pouw (2024), which used clips of the videos as input. Where the static model was a binary model, the dynamic model first classified every video as gesture or no gesture and then classified the gesture category as either palm up open hand (positive) or not (negative). The goal of this comparison was to gain an understanding of the advantages and challenges of each of the models and to see where opportunities for further research would lie.

To test this, I compared the performances of the models in a set of different conditions. For the positive cases, which were all some form of the palm up open hand gesture, I varied how many different forms of the gesture I included in the dataset. While more variation might make it more difficult for the model to classify all instances of the gesture as part of the same class, including all variation that was originally present in the dataset is more similar to real life gesturing (Lutgen et al. 2026). Moreover, including all variations in form would keep the class imbalance to a minimum as, for my original experiment, I did not adjust the size of the negative class to be the same

as the size of the positive class. Because training and testing the static model took less than a minute I was able to test six different combinations for this model. The dynamic model took longer to train and test, therefore I only tested two combinations for this model. In a follow-up experiment, I also tested the same conditions I used for the dynamic model again in the static model but with a set number of training and test instances to control for sample size.

For the negative cases, which were all other types of (non-)gestures, I only created two conditions, based on whether or not the *Unsure* subclass was included, This subclass contained all gestures that looked like a palm up open hand gesture in form, but were another type of gesture according to my intuitions as native speaker of Dutch. While including them would be most similar to real life, there would be no way for the models to know these gestures were not part of the positive class without having more context, for example by including information on what is being said by the participants, how they said it, or the (hand) movements preceding and following the *Unsure* gesture. By combining the two variables I ended up with a total of twelve conditions for the static model and four conditions for the dynamic model.

I evaluated the performance of the two models using precision, recall, accuracy and F1 score, and by studying their confusion matrices. I also performed a qualitative analysis of a subset of samples in each category in the confusion matrix. For the static model, I performed an additional error analysis, looking into an additional sample from the positive and negative cases to study how each of the subclasses was handled by the model.

When comparing the evaluation metrics of the original two experiments, I found that the static model performs better than the dynamic model. The static model performs best in the conditions with more variation, possibly because it is more sensitive to the class imbalance in the other conditions. For the dynamic model, the opposite is true: here, the increase in variation within the positive cases creates more challenges for the model than the class imbalance. In the follow-up experiment for the static model I trained and tested the model again but with a set sample size for both the positive and negative cases. Even though this led to a decrease in the amount of training instances, this still yielded an even better performance than was originally the case.

When comparing the models, another factor that should be taken into account is the cost of training and running the model. The static model was significantly faster to run than the dynamic model. However, there was a lot more manual labor involved in the data preprocessing because I needed to take screenshots of the hold phase of each of the gestures. While the dynamic model took longer to run, there was less manual labor involved as the video clip segmentation was done automatically based on the annotations.

My findings support the idea of Cienki (2021) that different forms of palm up open hand gestures are points on a continuum. The models used in this thesis were able to generalize across these different forms based on shared features, and were able to classify all forms correctly, as is shown in the error analyses I performed.

To conclude, I will answer my research question as follows: while gesture recognition models are able to recognize palm up open hand gestures, a lot more research is needed to create reliable models with high performance scores. In this research, it is important to always focus on the variation that is present in real life gesturing and how this is handled by the models that are used, as this variation is an essential part of language as it is used by both individuals and groups to express themselves and shape their

identity.

6.2 Future Work

Because this thesis focused on exploratory work, the results of this study show several interesting opportunities for future research. Here, I will discuss some options for future work based on this thesis.

6.2.1 Suggestions for Both Models

Firstly, I would like to discuss some suggestions for further research that are relevant for both the static and the dynamic model.

Both models could benefit from more training data. While studies with limited data are valuable, both because this minimizes training time and because for some languages more data is simply not available, studying how the models' performances are affected by this is also interesting. Training and testing the models with varying amounts of data could show to what extent the models are affected by this limited availability of data.

Furthermore, a study where videos are taken from different angles could show the effect of changing the camera angle. Both a study where the training data is taken from a different angle than the test data, and a study where there is a mix of angles present in both training and test sets would provide interesting perspectives on the functioning of the gesture recognition models.

Additionally, for both static and dynamic gesture recognition models, it would be interesting to compare more models. While the models used here can give an indication of initial advantages and challenges, there is a large variety of gesture recognition models available that could be adapted to recognize palm up open hand gestures. Comparing their performance might lead to new insights and better results.

Finally, due to the fact that both models seemed to struggle differentiating between an open hand that was facing upwards and an open hand that was facing down, it would be very valuable to look into how exactly this rotation influences the models' ability to recognize palm up open hand gestures. By carefully creating a data set with videos or images of open hand gestures that vary in the direction the palm is facing, we could see with more certainty how the models handle these cases.

6.2.2 Static Model

Secondly, I will discuss some suggestions for future work on the static gesture recognition model. One of the major disadvantages of this model is that there is a lot of manual labor involved in selecting the hold of each gesture and taking a screenshot to provide as input for the gesture recognition model. However, it would be interesting to see if this model could be combined with another model that focuses on recognizing the hold of a gesture, or potentially just classifies in which frames there is movement. While this might increase the computational cost of the gesture recognition pipeline as a whole, it would decrease the amount of manual labor involved.

Additionally, it would be interesting to see how the performance of the static model changes if not only the joints in the hand, but also the arm and shoulder were used as input to the model. As I explained, the palm up open hand gesture can involve movement of not only the hand, but also of the arm and shoulder. However, it might

be the case that the performance of the static model is less affected by the different form variants because it bases the assigned label only on features that reflect that shape of the hand. Future research could show if including the arm and shoulder would positively or negatively impact the performance.

Finally, a suggestion for future work is to train the static model on data that is controlled for participant sample size. As the per participant analysis showed, the static model is sensitive to the number of gestures made by a participant in the data set. To further improve the model's overall performance and to make it more robust, I propose to train the model with a set number of positive and negative gestures per participant.

6.2.3 Dynamic Model

The dynamic model could be studied further in several ways. Because this model seemed very sensitive to variation in the positive class, it would be interesting to see what the effect would be of training and testing this model in a purely artificial setting, with gesture data created by participants who were asked to specifically produce a palm up open hand gesture several times. This could show whether the performance of the model is indeed affected by the variation in form.

6.2.4 Other Suggestions

Finally, it would also be interesting to repeat this study for other recurrent gestures, potentially in other languages. While I did narrow down my research question to focus on palm up open hand gestures, due to the availability of this gesture in my dataset, the broad research question still is a valuable question to answer. Recurrent gestures are a relatively new domain within gesture studies, and seeing if the techniques used in this thesis are relevant for other types of recurrent gestures could make quantitative analyses of recurrent gestures a much simpler and faster process.

Appendix

Appendix A

Consent Forms

Informatiebrief

Titel van het onderzoeksproject

Onderzoek naar communicatie tussen twee participanten

Dit document heeft betrekking op onderzoek binnen de School der Geesteswetenschappen aan de Vrije Universiteit Amsterdam (VU) waarbij deelname door of data van personen een rol speelt. De VU hecht veel waarde aan het waarborgen van uw privacy en aan de bescherming van uw persoonlijke gegevens. In deze informatiebrief bespreken wij hoe we uw persoonlijke gegevens verwerken in overeenstemming met de Algemene Verordening Gegevensbescherming (AVG) en de Uitvoeringswet Algemene Verordening Gegevensbescherming (UAVG).

1. Wat is het doel van het onderzoek?

Het doel van dit onderzoek is meer leren over communicatie in natuurlijke gesprekken in het Nederlands.

2. Wat wordt er van u verwacht?

Tijdens dit onderzoek zal u samen met een andere deelnemer een gesprek voeren. Om te zorgen dat alle duo's over hetzelfde onderwerp praten zal u eerst een korte video kijken. Daarna zullen een voor een enkele vragen getoond worden op een laptop. U wordt gevraagd elk van deze vragen voor een paar minuten te bespreken en dan zelf door te gaan naar de volgende vraag. Na ongeveer 20 minuten hebben wij genoeg data verzameld en zullen wij het onderzoek beëindigen. Uw deelname aan dit onderzoek zal in zijn geheel ongeveer 30 minuten duren. U kan op elk moment uw deelname aan dit onderzoek beeindigen zonder hier een reden voor te verschaffen.

3. Beloning

Voor uw deelname aan dit onderzoek wordt u gecompenseerd met een VVV cadeaucaart ter waarde van 15 euro.

4. Wat gebeurt er met uw gegevens?

Voor dit onderzoek verzamelen wij verschillende gegevens: video en audio opnames van jullie gesprek. Dit doen wij om een zo compleet mogelijk beeld te krijgen van jullie communicatie. Conform de richtlijnen van de AVG zullen wij niet meer persoonlijke informatie verzamelen dan strikt noodzakelijk. Wij hechten veel waarde aan uw privacy. Daarom zullen wij uw persoonlijke gegevens op de volgende manier beveiligen.

Figure A.1: Image of the participant information document that was given before the experiment to inform the participants about the procedure.

- Alle data zal gedurende het onderzoeksproject opgeslagen worden in Yoda, een cloudopslag repository voor wetenschappelijke integriteit. Na het onderzoeksproject zal alle data hier voor minstens 10 jaar bewaard worden.
- Na uw deelname aan dit onderzoek zal u een formulier krijgen waarin u aan kan geven waar en hoe uw data gebruikt mag worden. Hier kan u aangeven of en waar data waarop u herkenbaar in beeld of te horen bent gepubliceerd mag worden. Hetzelfde kunt u doen voor data die niet tot u terug te leiden is zoals transcipties van de audio. U kunt ook aangeven dat u helemaal niet wil dat er data gepubliceerd wordt. Alle data die niet gepubliceerd mag worden zal na 10 jaar uit Yoda verwijderd worden.
- De resultaten die voortkomen uit dit onderzoek kunnen gebruikt worden voor wetenschappelijke doeleinden, zoals publicaties, congrespresentaties, etc.

5. Wie heeft er toegang tot uw gegevens?

Sommige mensen mogen uw persoonsgegevens inzien. Dit is om te controleren of het onderzoek goed en betrouwbaar is. Mensen die uw gegevens kunnen inzien zijn dr. Lucia Donatelli, prof. dr. Alan Cienki en Fenna Berger. Als u de toestemmingsverklaring ondertekent, geeft u toestemming voor het verzamelen, bewaren en inzien van uw persoonsgegevens door deze personen.

6. Mogelijke voor- en nadelen

U heeft zelf waarschijnlijk geen direct voordeel van deelname aan dit onderzoek. Uw deelname kan wel bijdragen aan meer kennis over communicatie. Een nadeel dat u van dit onderzoek kan ervaren is de tijd die het u kost. Voor dit onderzoek gaat het om een tijdsinvestering van 30 minuten.

7. Rechten en ethische toetsing

Deelname aan het onderzoek geschiedt op vrijwillige basis. Conform de richtlijnen van de AVG heeft u onder bepaalde voorwaarden het recht om uw persoonlijke gegevens in te zien, te corrigeren of te laten verwijderen. Als u tijdens het onderzoek besluit om uw medewerking te staken, zal dat op geen enkele wijze gevolgen voor u hebben. Echter, de gegevens die u hebt verstrekt tot aan het moment waarop uw deelname stopt, zullen in het onderzoek worden gebruikt.

Indien u van deze rechten gebruik wilt maken dan kunt u contact opnemen met de functionaris gegevensbescherming van de Vrije Universiteit (zie contactgegevens hieronder). Heeft u klachten? Dan kunt u zich in eerste instantie wenden tot de onderzoeker (zie contactgegevens hieronder). Is uw klacht daarmee niet opgelost, dan kunt u een klacht indienen bij de functionaris gegevensbescherming, en/of de Autoriteit Persoonsgegevens (zie contactgegevens hieronder).

De Ethische Toetsingscommissie Onderzoek van de School der Geesteswetenschappen van de Vrije Universiteit heeft dit onderzoek bekeken en van advies voorzien op 23/4/2026 onder referentienummer ETCO26.17.

8. Heeft u vragen?

Bij vragen kunt u contact opnemen met:

Naam:	Fenna Berger
E-mailadres:	f.berger@student.vu.nl
Telefoonnummer:	0637471104

Ik heb de informatiebrief gelezen en ik ben in de gelegenheid gesteld om vragen te stellen:

☐ ja ☐ nee

Deelnemer

_____	_____	
Naam	Handdtekening	Datum

Onderzoeker

_____	_____	
Naam	Handdtekening	Datum

Contactgegevens van de onderzoeker

Naam:	Fenna Berger
E-mailadres:	f.berger@student.vu.nl
Telefoonnummer:	0637471104

Contactgegevens van de functionaris gegevensbescherming (Vrije Universiteit)

Vrije Universiteit Amsterdam
Functionaris voor Gegevensbescherming
De Boelelaan 1105
1081 HV AMSTERDAM
functionarisgegevensbescherming@vu.nl

Contactgegevens van de Autoriteit Persoonsgegevens

Autoriteit Persoonsgegevens
PO Box 93374
2509 AJ DEN HAAG
(+31)-(0)70-8888500



Informed Consent Formulier

Titel van het Project

Onderzoek naar communicatie tussen twee participanten

Voorwaarden

- Ik heb de informatie over het project in het document met participanteninformatie gedateerd op 23/4/2026 gelezen en begrepen.
- Ik heb de mogelijkheid gehad om vragen te stellen over dit project en mijn deelname.
- Ik stem vrijwillig toe met deelname aan dit project.
- Ik begrijp dat ik voor mijn deelname aan dit project vergoed zal worden met een VVV cadeaukaart ter waarde van 15 euro.
- Ik begrijp dat ik mijn deelname aan dit project op elke moment kan beeindigen zonder opgaaf van redenen.
- Ik begrijp dat de onderzoeker audio- en videoopnames zal maken van onze interactie. Ik begrijp dat de data opgeslagen zal worden op een beveiligde drive en vervolgens gedeeld zal worden met het supervisieteam.
- Ik begrijp dat de analyses en de uiteindelijke resultaten zoals beschreven in het document met participanteninformatie anoniem zullen worden vormgegeven. De onderzoeker kan bevindingen presenteren op conferenties of in publicaties maar mijn identiteit zal niet bekend worden gemaakt in gepubliceerde verslagen.
- Ik begrijp dat de data opgeslagen zal worden in Yoda, een cloudopslag repository voor wetenschappelijke integriteit. Ik begrijp dat andere wetenschappers alleen toegang krijgen tot deze data als zij erin mee instemmen de van de data te waarborgen.
- Ik begrijp dat de onderzoeker van dit project ermee instemt dat deze studie voldoet aan de ethische richtlijnen van de Faculteit der Sociale en Geesteswetenschappen van de Vrije Universiteit (VU). Ik begrijp dat ik als ik een klacht heb contact op moet nemen met de Functionaris Gegevensbescherming van de VU (zie contactinformatie hieronder) en/of de Autoriteit Persoonsgegevens (AP) (zie contactinformatie hieronder).

Ik stem in met deelname aan dit onderzoek: ☐ ja ☐ nee

Participant

_____	_____	_____
Naam	Handtekening	Datum

Onderzoeker

_____	_____	_____

Figure A.2: Image of the consent form that was given to the participants before the experiment.

VU VRIJE UNIVERSITEIT AMSTERDAM | FACULTY OF HUMANITIES

Naam	Handtekening	Datum

Contactgegevens van de onderzoeker

Naam:	Fenna Berger
E-mailadres:	f.berger@student.vu.nl
Telefoonnummer:	0637471104

Contactgegevens van de Functionaris Gegevensbescherming (VU)

Vrije Universiteit Amsterdam
Data Protection Officer
De Boelelaan 1105
1081 HV AMSTERDAM
functionarisgegevensbescherming@vu.nl

Contactgegevens van de Autoriteit Persoonsgegevens (AP)

Autoriteit Persoonsgegevens
PO Box 93374
2509 AJ DEN HAAG
(+31)-(0)70-8888500

Participanten ID ___

Uw instemming met het verwerken en gebruik van de data

Met uw instemming zal de gemaakte video gebruikt worden om onderzoek te doen naar communicatie. U heeft het recht om het gemaakte videomateriaal te bekijken.

Geef hieronder aan hoe het videomateriaal van uw conversatie gebruikt kan worden. De data zal alleen gebruikt worden op de manier(en) waar u toestemming voor heeft gegeven. Geef hieronder alstublieft aan waar u toestemming voor geeft.

1. Anoniem gemaakte informatie kan gebruikt worden in wetenschappelijke publicaties en presentaties (bijvoorbeeld een transcriptie zonder mijn naam of een schematische weergave van mij)

   Ja of nee? _____

2. (Delen van) de audio en videoopnames die gemaakt zijn in dit onderzoek waarop ik herkenbaar te zien of te horen ben kunnen gebruikt worden in wetenschappelijke publicaties en presentaties (bijvoorbeeld in wetenschappelijke tijdschriften, voor onderwijsdoeleinden of in presentaties op academische conferenties)

   Ja of nee? _____

3. Delen van de audio en videoopnames van mijn deelname kunnen gepost worden op wetenschappelijke websites (bijvoorbeeld als demonstratie van een onderzoeksproject of als voorbeeld in een wetenschappelijk artikel)

   Ja of nee? _____

I give my consent to the use of materials with my participation as indicated above.

Datum:

Handtekening:

E-mail:

De onderzoeksgroep zal de verzamelde data alleen gebruiken in overeenkomst met de punten die door de participant ondertekend zijn.

Datum:

Handtekening: [van de onderzoeker]

Figure A.3: Image of the second part of the consent form that was given to the participants after completing the experiment.

Appendix B

Annotation Manual

For each composite video, I made a new project in ELAN (Max Planck Institute for Psycholinguistics, The Language Archive (2026)). In this project, I created four tiers, two for both participants' PUOH gestures (class 1) and two for both participants' other gestures or no gesture annotations (class 0). For all gestures I annotated the full range of the gesture, including the stroke, hold, and retraction. In cases where this was not possible, for example because the gesture was followed immediately by another gesture, I only annotated the stroke and hold. For the videos that were used as input for the [MPI] model, I used the full annotation. For the images that were used as input for the [insert other name] model, I selected the initial part of the hold phase of the gesture. All gestures where the active hand(s) was/were not fully visible were excluded.

I distinguished between four categories of class 1 gestures, adjusted from Cienki et al. (2025) as I described in Chapter **??**:

- Thumb lift (TL): the gesture includes only movement of the thumb(s);
- Hand turnout (HT): the gesture includes a rotation of the wrist;
- Open hand diagonal (OHD): the gesture includes a rotation and/or extension at the elbow;
- Palm up open hand (PUOH): the gesture includes a rotation and/or extension at the shoulder.

The class 0 is meant to contain any gesturing that is not a palm up open hand gesture as well as moments where the participant is not gesturing at all. In order to be able to select what type of non-palm up open hand segments I wanted to include in my data set in the experiment, I created four different subclasses:

- Other gesture (OG): any gesture that is not a form of the PUOH gesture;
- Self-adapter (SA): gestures such as fidgeting, scratching a body part or adjusting one's hair;
- No gesture (NG): parts of the data collection where the participant is not gesturing;
- Unsure (US): this class contains all gestures that look like a PUOH gesture in form, but, according to my intuitions as a native speaker of Dutch, were used in a different manner than is usually the case for a PUOH gesture (such as iconic).

The length of every NG label was roughly between 1 and 2 seconds, which was also the range in which most of the lengths' of the class 1 gestures fell. If possible, I tried to capture different resting positions for each participant. Because some participants were making self-adapter gestures throughout almost the whole video, I did not annotate all of them. Instead, I only annotated the self-adapter gestures that were not present throughout the entire recording.

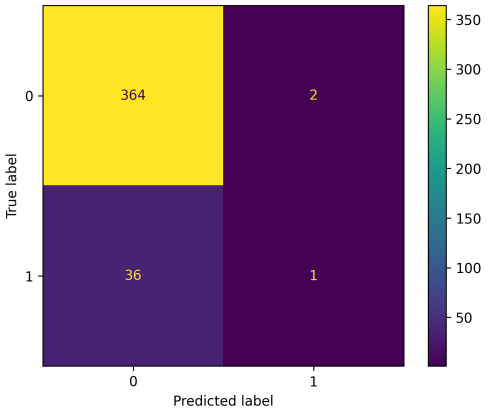
Every annotation was labeled with a name indicating:

- The video id;
- The gesture id: within the video I gave each gesture a number to ensure that no two gestures would have the same name;
- A 1 or a 0 indicating the class of the gesture;
- The subclass label;
- And an L or an R to indicate which participant made the gesture.

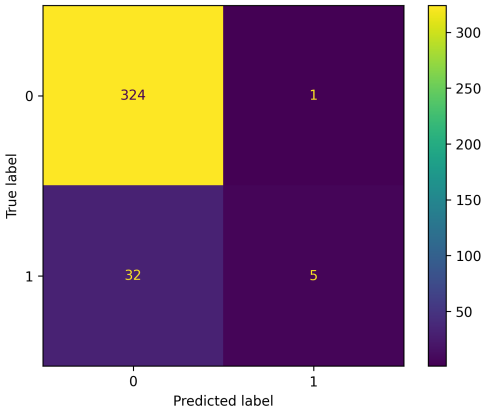
For example, the label *5-12-1-OHD-R* would refer to the twelfth gesture in video five that belongs to class 1, is an open hand diagonal and is made by the participant sitting on the right.

Appendix C

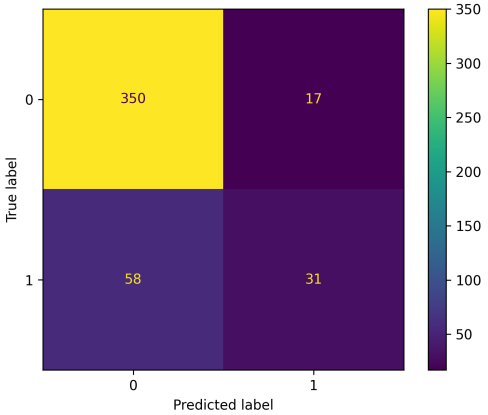
Confusion Matrices for the Image-based Model



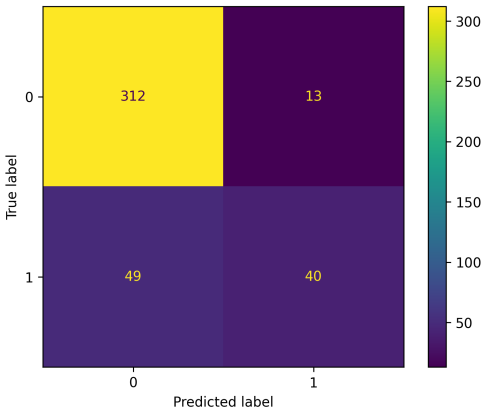
(a) HT+TL-only/With-US



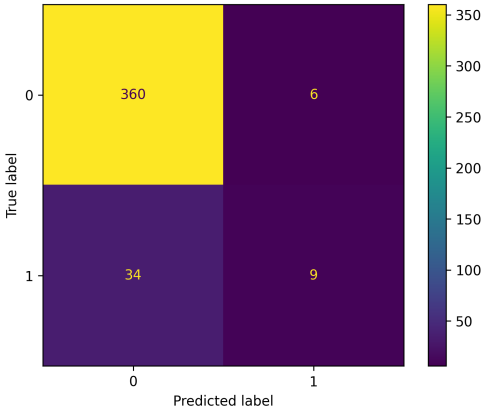
(b) HT+TL-only/Without-US



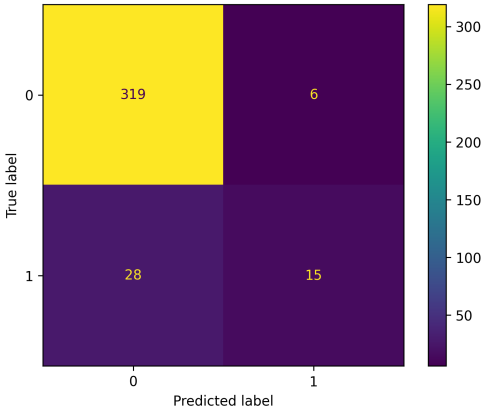
(c) OHD-only/With-US



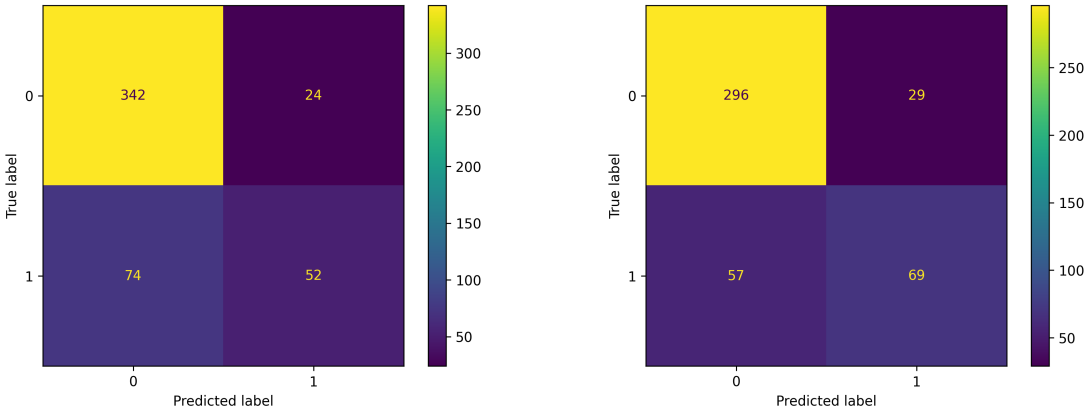
(d) OHD-only/Without-US



(e) PUOH-only/With-US

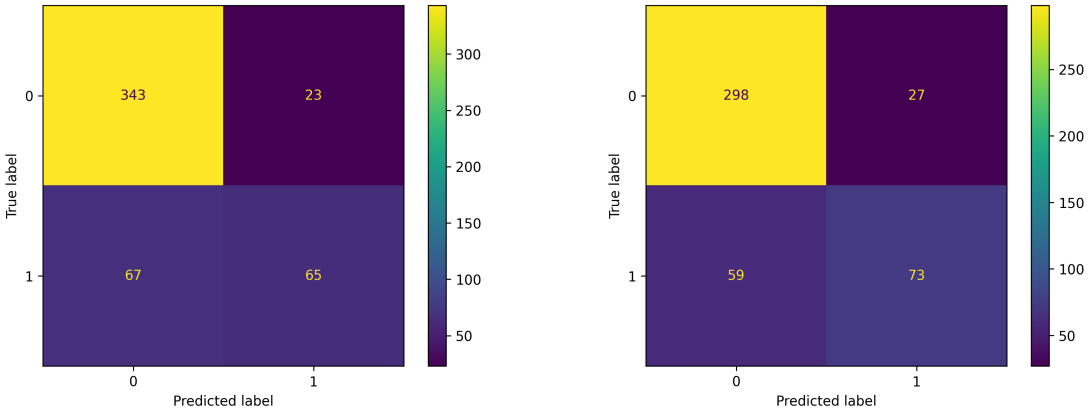


(f) PUOH-only/Without-US



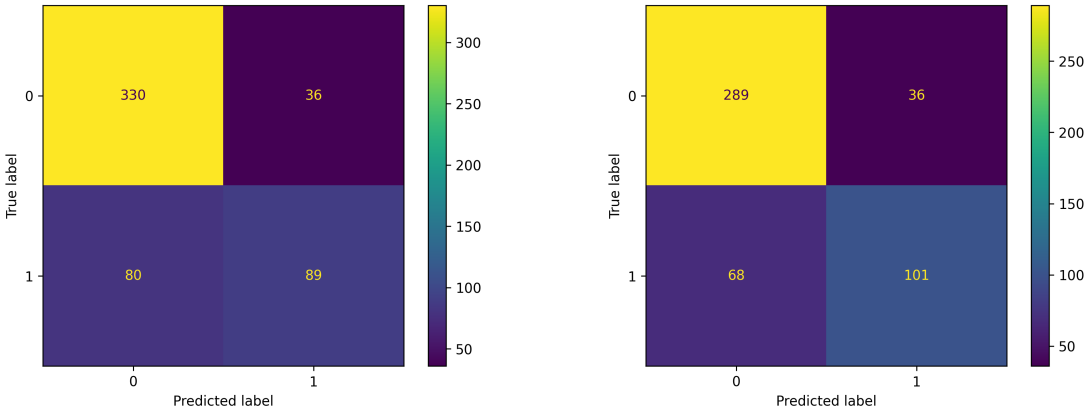
(a) OHD+HT+TL/With-US

(b) OHD+HT+TL/Without-US



(c) PUOH+OHD/With-US

(d) PUOH+OHD/Without-US



(e) All-Variants/With-US

(f) All-Variants/Without-US

Figure C.2: Confusion matrices for the static model.

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